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PLATING POSTERS INC · METAL FINISHING REFERENCE SERIES

BRIGHT NICKEL

THE COMPLETE TRAINING MANUAL

A full training course for the brand-new operator — from "what is an anode?" to a signed certificate of completion. Assumes you know nothing. Teaches you everything.

01 CLEAN ▸ 02 RINSE ▸ 03 ACTIVATE ▸ 04 RINSE

05 PLATE ▸ 06 RINSE ▸ 07 POST-TREAT ▸ 08 DRY

OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 · 2026 · BEGINNER-FRIENDLY STYLE

FIND YOUR WAY

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★ **REMEMBER**

This manual is a training and reference tool — **not** a substitute for your facility's written procedures, your supplier's TDS/SDS, or your supervisor's instructions. When this manual and your shop's official procedure disagree, **follow your shop's official procedure and ask your supervisor.**

PART ONE — FRONT MATTER & FUNDAMENTALS

START HERE

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU ARE IN EXACTLY THE RIGHT PLACE

Hello, and welcome to the team. If you've never plated a single part in your life — if you don't know an anode from an apron — you are in exactly the right place. This manual was written for you: the brand-new operator, the curious apprentice, and anybody who has ever looked at a tank line and thought, "What on earth is actually going on in there?"

Here's a promise we'll make right up front: **we are going to assume you know nothing.** That's not an insult — it's a teaching style. Every term gets defined the first time it shows up. Every step gets a reason attached to it ("do this *because*…"). We never wave our hands and say "you already know this." Nobody is born knowing what a "Watts bath" is. You'll learn it here, from zero.

By the time you finish this manual, you'll understand what electroplating is, why bright nickel is one of the most important finishes in all of manufacturing, what every tank on the line is *for*, why the tanks sit in the order they do, and — most importantly — how to work safely around chemicals and electricity that genuinely demand your respect. Let's get started.

• HOW TO USE THIS MANUAL

This manual is built to be read front-to-back the first time, then kept on the shelf as a reference you flip back into forever. It follows the plating line itself, in order, stage by stage. After this fundamentals section, you'll walk through all eight stages of the bright nickel process — Clean, Rinse, Activate, Rinse, Plate, Rinse, Post-Treat, and Final Rinse/Dry — one chapter at a time. Reading it in order matters, because (as you'll soon learn) **the order of the plating line is not random** — and neither is the order of this book. One housekeeping note: the process has **eight stages** but only seven numbered Station chapters. **Stage 8 (Final Rinse / Dry) is short, so we teach it at the end of the Station 07 chapter** — it still gets its own Teach-Back and sign-off, and the Training Matrix certifies it as its own row.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS

Throughout the manual you'll see callout boxes flagged with these labels. They're your shortcut to the good stuff.

💡 **TIP**
A pro shortcut or trick of the trade — the kind of thing a 20-year veteran would lean over and tell you. Things that make your day easier or your parts come out better.

★ **REMEMBER**
A key takeaway. If you forget everything else on a page, don't forget this. These are the load-bearing ideas.

⚠ **HEADS-UP**
Safety and danger. Read every single one of these, every time. We use this icon *liberally*, because in a plating shop safety isn't a separate chapter — it's baked into every task. When you see HEADS-UP, slow down and pay attention.

⚙ **TECHNICAL STUFF**
Optional deeper chemistry, for the curious. If you want the why-behind-the-why, dig in. If you just want to run good parts, you can skip these and lose nothing essential.

☀ **FUN FACT**
Light, memorable trivia — the stuff that makes plating genuinely interesting and gives you something to share at lunch.

💡 **TIP**
First time through, don't try to memorize every number. Read for *understanding* — what each stage does and why. The exact numbers live in the stage chapters and on the shop-floor posters right at the tanks, where you can glance at them while you work. Understanding sticks; memorized numbers fade. Get the concepts first.

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

THE WHOLE FOUNDATION — MASTER THIS AND EVERY STAGE CHAPTER CLICKS INTO PLACE

By the end of this fundamentals section, you will be able to:

1. **Explain what electroplating is** in plain language — and describe, in simple terms, how electricity moves metal from one place to another.
2. **Define bright nickel** — what makes it "bright," what it's used for, and why it's called "the workhorse of decorative plating."
3. **Describe how an electroplating line works** as a connected system, not just a row of separate tanks.
4. **Name all eight stages** of the bright nickel process and explain what each one accomplishes.
5. **Explain why the stage order matters** — why you can't skip a rinse, swap two steps, or "save time" by cutting a corner.
6. **List the required PPE** and describe the emergency/first-response actions for the hazards on a nickel line.
7. **Describe the overall safety philosophy** of working around hot alkalis, strong acids, nickel (a skin sensitizer and inhalation carcinogen), and high electrical current.

• THE EIGHT-STAGE PROCESS AT A GLANCE

Here is the whole bright nickel line in one strip. Don't memorize it yet — just get a feel for the rhythm: **Clean → Rinse → Activate → Rinse → Plate → Rinse → Post-Treat → Rinse/Dry.**

01	02	03	04	05	06	07	08
CLEAN	**RINSE**	**ACTIVATE**	**RINSE**	**PLATE**	**RINSE**	**POST-TREAT**	**RINSE/DRY**
140-180F	firewall	acid dip	guard	main event	+ bake?	chrome/seal	spot-free

★ **REMEMBER**

Notice the pattern: **a rinse follows almost every chemical step.** That's not wasted motion — rinses are the firewalls of the line. Every working chemistry wants to "hitch a ride" into the next tank on the surface of your part, and rinses are what stop that. Rinses aren't filler between the "real" steps; they're some of the most important steps you'll perform.

CHAPTER 1

WHAT IS ELECTROPLATING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT PLATING IS

THE ONE-SENTENCE VERSION

Electroplating is using electricity to coat one metal with a thin layer of a different metal. That's the whole idea. We take a part — say, a steel faucet handle — and use an electric current to grow a microscopically thin, even coat of nickel all over its surface. The nickel makes it shinier, more corrosion-resistant, and prettier than bare steel ever could be.

AN EVERYDAY ANALOGY

Imagine you're spray-painting a chair. You dip your brush in paint and coat the chair so the wood underneath is protected and looks nicer. Electroplating is a lot like that — except instead of paint we use *real metal*, and instead of a brush we use *electricity* to make the metal stick. And unlike paint, plated metal bonds to the part at a level so intimate it becomes, for all practical purposes, part of the surface itself.

HOW ELECTRICITY MOVES METAL: THE KITCHEN-SINK PICTURE

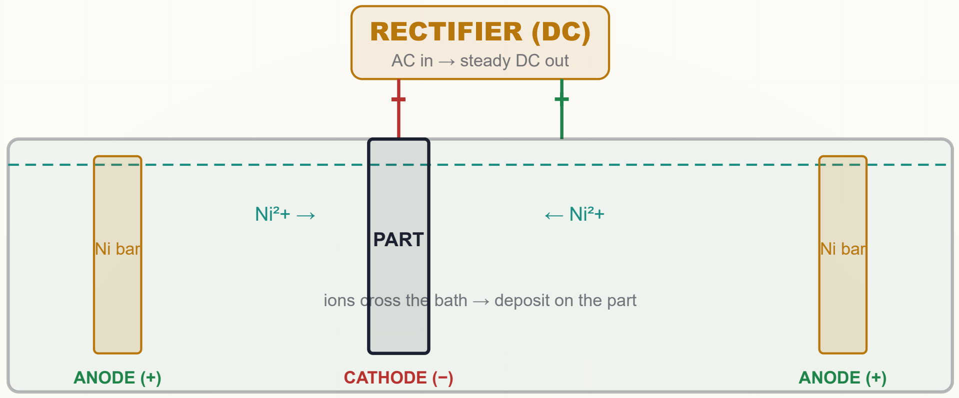
Here's the part that feels like magic the first time you hear it. Picture a tank full of a special liquid called the **plating bath** (or **electrolyte** — a fancy word for "a liquid that conducts electricity because it has dissolved chemicals in it"). For bright nickel, this liquid is loaded with dissolved nickel, floating around as tiny, invisible, electrically-charged particles called **ions**. (An **ion** is just an atom that has gained or lost some electrical charge — for nickel, each ion carries a positive charge, written Ni^{2+}.)

Now we put two things into that tank and hook them up to a power supply called a **rectifier** (a device that turns ordinary wall electricity into the steady, one-direction DC current that plating needs):

- **The cathode** — this is your *part*, the thing you want to coat. It's connected to the **negative** terminal.
- **The anode** — this is a bar or basket of pure nickel metal. It's connected to the **positive** terminal.

When you turn on the current, here's the beautiful chain reaction:

1. The positively-charged nickel ions (Ni^{2+}) floating in the liquid are attracted to your negatively-charged part (opposites attract!). They swim over to it.
2. At the surface of the part, each nickel ion grabs electrons from the current and turns back into solid metal nickel, sticking firmly to the part. Atom by atom, a nickel coating *grows* right on the surface.
3. Meanwhile, over at the anode, the solid nickel bar slowly *dissolves* into the liquid, replacing the ions that just plated out — so the bath doesn't run out of nickel.



★ **REMEMBER**

Electroplating is a transfer system: nickel leaves the anode (the metal bar), travels through the liquid as ions, and re-forms as solid metal on the cathode (your part). The rectifier provides the electrical "push." **Negative = your part. Positive = the metal anode.**

THE CHEMISTRY, THE HISTORY, AND THE HAZARD

TECHNICAL STUFF

The reaction at your part (cathode) is a **reduction**: $Ni^{2+} + 2e^{-} \rightarrow Ni^{0}$ (solid). At the anode it's the reverse — an **oxidation**: $Ni^{0} \rightarrow Ni^{2+} + 2e^{-}$. The whole process obeys **Faraday's laws of electrolysis**: the amount of metal deposited is directly proportional to the electric charge (amp-hours) pushed through. This is why operators track **amp-hours (Ah)** — current × time — as the true measure of "how much plating happened."

FUN FACT

Electroplating is older than the light bulb and the telephone. The first practical commercial plating patents date to around 1840 in England, when the Elkington cousins commercialized silver and gold electroplating. Fancy restaurant dinner forks have been getting electroplated for nearly two centuries.

HEADS-UP

That "wall electricity turned into DC" is no joke. Plating rectifiers push enormous current — hundreds or even thousands of amps. It is low voltage but very high current, and the tank is full of conductive liquid. **Never reach into an energized tank, and never perform maintenance until equipment is locked out and tagged out (LOTO).** Electricity plus saltwater plus your hand is a combination that demands total respect.

CHAPTER 2

WHAT IS BRIGHT NICKEL?

AND WHY DO WE USE IT — THE WORKHORSE OF DECORATIVE PLATING

PLAIN NICKEL VS. BRIGHT NICKEL

If you plated a part in a basic nickel bath with nothing fancy added, you'd get a coating that works fine but looks kind of… blah. Matte. Dull. A little yellowish. Functional, but not pretty. **Bright nickel** is different. It comes out of the tank looking like a mirror — brilliant, reflective, smooth — *with no polishing or buffing afterward.* You pull the part out and it's already gorgeous. So what's the secret? It's not the nickel itself.

FUN FACT

The "bright" in bright nickel doesn't come from the nickel — it comes from tiny amounts of special organic chemicals called **brighteners** added to the bath. Without them, the very same nickel bath gives a dull, slightly yellowish, matte deposit. The brighteners are the magic ingredient. We'll meet them properly in the Plating chapter.

LEVELING: THE OTHER SUPERPOWER

Bright nickel doesn't just make things shiny — it also does something called **leveling.** Leveling means the nickel preferentially fills in tiny scratches, machining marks, and microscopic valleys on the surface, smoothing them out as it plates. The result is a surface that's actually *smoother* than the metal you started with — a bit like spackling a wall before painting, except it happens automatically as the metal deposits.

TECHNICAL STUFF

Leveling works because certain brightener molecules (called **levelers**, or Class II brighteners) cling more tightly to the high points (peaks) of the rough surface, where current density is highest. By partially blocking deposition on the peaks, they let metal build up faster in the low points (valleys), gradually evening out the surface. It's a self-correcting geometry trick driven by where the electric field is strongest.

WHERE BRIGHT NICKEL IS USED (YOU'VE TOUCHED IT TODAY)

Bright nickel is *everywhere*. By tonnage, it's the most-used nickel plating process in the world. You almost certainly handled something plated with it today without realizing it:

- **Automotive trim** — grilles, badges, door handles, wheel accents
- **Plumbing fixtures** — faucets, shower heads, towel bars (that shiny "chrome" bathroom hardware? Bright nickel is underneath it)
- **Hardware** — drawer pulls, hinges, tools, fasteners
- **Appliances and furniture** — decorative legs, frames, knobs
- **Anything** that needs to look like polished metal and resist corrosion

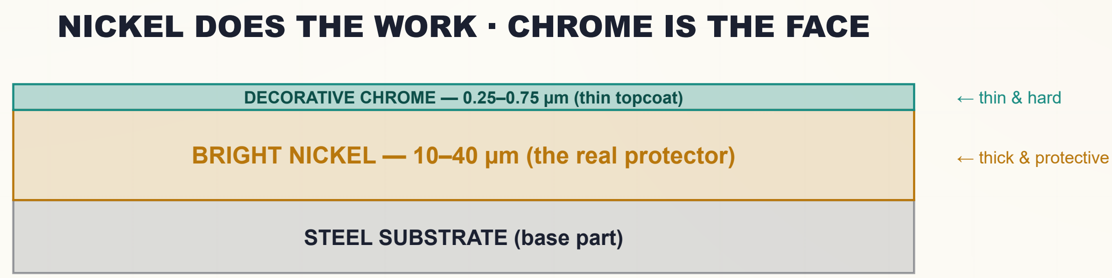
It plates onto a wide range of base metals (called **substrates** — the material your part is made of before plating): steel, brass, copper alloys, zinc die cast, and — with extra preparation — even aluminum and plastics.

THE BIG SECRET: NICKEL IS THE REAL PROTECTOR, NOT CHROME

Here's something that surprises almost every new operator. That brilliant "chrome" finish on a car bumper or a faucet? **The chrome layer is incredibly thin — only about 0.25 to 0.75 micrometers** (a micrometer, written µm, is one-thousandth of a millimeter — far thinner than a human hair). The chrome is there for its blue-white color, hardness, and tarnish resistance.

The *actual corrosion protection* — the layer doing the real work of keeping rust away — is the **bright nickel underneath**, which is much thicker, typically **10 to 40 µm** (and 20–40 µm under automotive chrome). On a typical decorative part, nickel does the heavy lifting; chrome is just the pretty, durable topcoat.

THE FINISH, LAYER BY LAYER (NOT TO SCALE)



REMEMBER

In a nickel-chrome decorative finish, **bright nickel is the primary corrosion barrier — not the chrome.** The chrome is thin (0.25–0.75 µm) and exists mainly for appearance and hardness. The nickel (10–40 µm) is what actually protects the part. This is why over 90% of all decorative chrome in the world is plated *on top of* bright nickel.

WHY OPERATORS CALL IT "THE WORKHORSE"

- It comes out **mirror-bright straight from the tank** — no buffing labor needed.
- It **levels** out minor surface flaws, so even imperfect parts can look great.
- It's **highly efficient** — typically around 95% (roughly 92–97%, supplier- and condition-dependent) of the current actually deposits metal (this is called **cathode efficiency**; the rest makes a little hydrogen gas, which matters for safety).
- It's the **foundation for decorative chrome** and works on a huge range of substrates.
- It has **over 80 years of industrial history** — extremely well understood, with tons of supplier support.

TIP

When someone says "we're running bright nickel," they're telling you to expect a *demanding, unforgiving* finish. A mirror shows everything — every fingerprint, every speck of dirt, every cleaning mistake. As the old shop saying goes: *"Nickel doesn't forgive dirty prep or lazy chemistry. Earn the brightness."* Most defects on a bright nickel line trace back to bad cleaning, not bad plating.

TECHNICAL STUFF

Bright nickel deposits contain a small amount of co-deposited **sulfur** (about 0.04–0.1% by weight), from the Class I brightener chemistry. This sulfur makes the deposit electrochemically "active." In advanced multi-layer systems (**duplex** or **triplex** nickel), a sulfur-free **semi-bright nickel** is plated first, then bright nickel on top. The two layers have slightly different electrical potentials, which cleverly forces corrosion to spread *sideways* along the boundary instead of drilling straight down to the steel — dramatically extending the finish's life. One of the most elegant tricks in all of metal finishing.

CHAPTER 3

HOW A PLATING LINE WORKS

THE BIG PICTURE — A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

A LINE IS A SEQUENCE OF TANKS

A plating “line” is exactly what it sounds like: a row of tanks, each holding a different chemical solution, arranged in a specific order. A part travels down the line — lowered into one tank, lifted out, moved to the next, lowered in again — until it comes out the far end fully finished. Parts ride down the line in one of two ways:

- **Rack plating** — Parts are individually hung on a **rack** (a frame with hooks or clips that holds parts and carries electrical current to them). Used for larger parts, or where appearance and precise placement matter. Most decorative bright nickel is rack-plated.
- **Barrel plating** — Lots of small parts (screws, washers, small fittings) are dumped together into a perforated rotating **barrel** that tumbles them through the chemistry. Far more efficient for high volumes of tiny parts, with slightly less control over each piece.

TIP

You’ll see different numbers for the same chemistry depending on rack vs. barrel — barrel baths often run a bit higher in nickel and chloride to make up for the lower current density tumbling parts receive. When in doubt about which numbers apply, check whether you’re running rack or barrel and read the matching column on your shop-floor poster.

THE WHOLE BRIGHT NICKEL SEQUENCE: 8 STAGES

Here is the complete process. Don’t memorize the details — just get a feel for the *flow.* Each stage gets its own full chapter later.

#	STAGE	WHAT IT DOES	ROUGH TEMP	ROUGH TIME
1	Alkaline Soak Clean / Electroclean	Strips oil, grease, polishing compound, shop dirt	140–180°F	3–10 min
2	Rinse (Pre-Activation)	Washes off cleaner before the acid step	Ambient	30–60 s
3	Acid Activation (Acid Dip)	Dissolves the invisible oxide; wakes up bare metal	Amb–110°F	15–60 s
4	Rinse (Pre-Plate)	The critical guard rinse — protects the nickel bath	Ambient	30–60 s
5	Bright Nickel Plating	The main event — deposits bright, level nickel	130–150°F	8–45 min
6	Rinse (Post-Plate)	Washes off nickel drag-out; protects downstream tanks	Ambient	30–60 s
7	Post Treatment (Chrome / Seal / Lacquer)	Adds the protective/decorative topcoat	95–125°F	3–8 min
8	Final Rinse / Dry	Clean DI rinse and dry — no water spots	Amb–140°F	30–60 s + dry

The basic rhythm is easy to remember: **Clean → Rinse → Activate → Rinse → Plate → Rinse → Post-Treat → Rinse/Dry.**

A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

★ **REMEMBER**

Every stage either protects the next stage or poisons it. There is no neutral step. A part that leaves the cleaner still carrying a film of cleaner will contaminate the acid. A part that leaves the acid still wet with acid will contaminate the nickel. What you do (or fail to do) at Stage 1 shows up as a defect at Stage 5. The line is a chain, and a chain is only as strong as its weakest link.

This is also why **the nickel bath itself is so precious.** Unlike the cleaner and the acid (dumped and remade regularly), a well-maintained bright nickel bath can run for *years — even decades* — without being thrown out. It's expensive, mature, and tuned. But here's the catch: many contaminants that sneak into it (dissolved copper, iron, zinc, chromium) **stay in it essentially forever** and are difficult or impossible to fully remove. Every milligram of junk you let drag into that tank is a permanent guest.

💡 **TIP**

Think of the bright nickel tank as the crown jewel of the line. Everything *before* it (clean, rinse, activate, rinse) exists largely to ensure only a perfectly clean, bare, well-rinsed part ever touches it. Everything *after* it (rinse, post-treat, rinse/dry) exists to protect and finish what the nickel created. The whole line orbits that one tank.

CHAPTER 4

WHY THE ORDER MATTERS

WALKING THE LINE — EACH STAGE SETS UP THE NEXT, AND THE SEQUENCE IS LOCKED BY CHEMISTRY

You might think the stage order is just convention — that you could shuffle steps or skip one to save time. You can't. Each stage *sets up* the next. Let's walk it together and see *why* each step has to be exactly where it is.

STAGE 1 — CLEAN (SOAK + ELECTROCLEAN): GET RID OF THE JUNK FIRST

Your incoming part is dirty. It's coated in machining oils, drawing lubricants, polishing compounds, fingerprints, and shop grime — some of it invisible. **None of that can be allowed near the plating bath**, because plating only works where the metal is truly clean and exposed. Plate over a film of oil and the nickel won't stick — you'll get peeling, skip plating (bare spots), or pitting. So cleaning comes first, because **everything downstream assumes the part is clean.** There are usually two cleaning steps: a **soak clean** (hot alkaline cleaner that lifts bulk oils) and an **electroclean** (cleaned with electric current that generates scrubbing gas bubbles to blast off the last microscopic film).

💡 **TIP**

The simplest, most powerful quality check in the whole shop is the **water-break test**, and it's free. Pull a cleaned part and watch how water sheets off it. A smooth, continuous, unbroken sheet ("zero seconds" to break) means the surface is clean. If water beads up or "breaks" into droplets, there's still oil there — *stop and re-clean.*

⚠ **HEADS-UP**

The cleaner is **hot (140–180°F) and strongly alkaline (pH 11–13.5)** — it causes chemical burns and can seriously injure your eyes. Electrocleaning runs **current through a conductive tank** and generates flammable hydrogen and oxygen gas. Wear full PPE, keep open flames away, ensure ventilation, and **lock out the rectifier before reaching near the tank.**

STAGE 2 — RINSE (PRE-ACTIVATION): DON'T CARRY THE CLEANER FORWARD

After cleaning, the part is wet with alkaline cleaner. The next step (Stage 3) is an *acid.* Acids and alkalis neutralize each other — drag a load of cleaner into the acid and you waste your acid neutralizing it instead of doing its real job. Worse, cleaner contains surfactants, silicates, and sodium that have no business going downstream toward the nickel bath. So we rinse. **This rinse is a firewall.**

★ **REMEMBER**

"Drag cleaner into acid and you waste acid. Drag cleaner into nickel and you poison it." Rinses exist to keep each chemistry from invading the next tank. The cleaner-to-acid rinse is your first firewall on the line.

STAGE 3 — ACID ACTIVATION (ACID DIP): WAKE UP THE METAL

The moment clean metal hits air and water, it grows an **invisible oxide layer** — a microscopically thin skin of metal-oxide (essentially a flash of tarnish/rust forming instantly). Nickel won't bond well to that oxide skin; it needs to grip *bare, active metal.* The acid dip dissolves that oxide off, exposing chemically "active" fresh metal — primed and ready for nickel. This is why it's called **activation.**

- **Under-dip** (too short/weak) = oxide remains = nickel peels off later (adhesion failure).
- **Over-dip** (too long/strong/hot) = you start etching and pitting the part itself, and load it with hydrogen.

Activation has to come *right before* plating, with only a rinse in between — because if freshly activated bare metal sits exposed, it just grows a new oxide skin and you're back where you started.

⚠ **HEADS-UP**

The acid dip uses **strong mineral acid** — typically HCl at 10–50% or H_2SO_4 at 5–20%. These cause severe burns and eye damage, and HCl gives off corrosive, choking fumes even at room temperature. The acid eating into metal also releases **flammable hydrogen gas.** Some die-cast activators contain **hydrofluoric acid (HF)** — in a lethal danger class of its own. HF penetrates skin painlessly, attacks bone, and disrupts your body's calcium in a way that can stop your heart; a seemingly minor splash can be fatal. **You must be trained on the HF first-aid protocol before going near it, calcium gluconate gel must be on hand and within reach, and any HF exposure is a call-911-immediately event even if it looks minor.** Full acid PPE, good ventilation, and **always add acid to water, never water to acid.**

⚙ **TECHNICAL STUFF**

Hydrogen embrittlement is a real, serious risk. During pickling and plating, atomic hydrogen can diffuse into steel's crystal structure. In high-strength steel (above ~40 HRC, or 180 ksi tensile), trapped hydrogen can cause catastrophic cracking later, under load — sometimes hours or days after the part is in service. The fix is **baking**: 375 ±25°F for at least 8 hours, started **within 4 hours** of the last plating step (ASTM B850). For prone parts, use *inhibited* acids and minimum dip time. A failed bolt or aircraft part can hurt people — always check the customer spec.

STAGE 4 — RINSE (PRE-PLATE): THE MOST IMPORTANT RINSE ON THE LINE

This rinse protects the crown jewel. The nickel bath runs for years and accumulates contaminants permanently. The part leaving the acid dip is wet with acid that now contains dissolved metals (iron from steel, copper from brass, zinc from die cast). If that drag-in reaches the nickel tank, those metals become permanent contaminants that wreck brightness and cause haze, streaks, and pitting. This is why this rinse targets a tighter conductivity (**< 200 µS/cm**) and why double rinses are strongly recommended.

★ **REMEMBER**

"Every contaminant you allow through this rinse becomes a permanent guest in the nickel." The pre-plate rinse is the single most important contamination barrier on the entire line. Drain well, rinse hard, and move the part quickly to the nickel tank before fresh oxide can form.

⚙ **TECHNICAL STUFF**

Conductivity (microsiemens per centimeter, µS/cm) is how operators quickly judge rinse cleanliness. Pure water barely conducts; dissolved chemicals make it conduct more. So *higher conductivity = more dragged-in chemistry = a dirtier rinse.* Watching conductivity is a cheap, instant way to know if your rinses are keeping up.

STAGE 5 — BRIGHT NICKEL PLATING: THE MAIN EVENT

Now — and only now, with a perfectly clean, freshly activated, well-rinsed part — does the actual nickel plating happen. This is the **Watts bath** (named after Oliver Watts, who developed the formula in the 1910s), containing:

- **Nickel sulfate ($NiSO_4 \cdot 6H_2O$), 240–330 g/L** — the main source of nickel metal.
- **Nickel chloride ($NiCl_2 \cdot 6H_2O$), 30–60 g/L** — helps anodes dissolve and improves how evenly the bath plates (“throwing power”).
- **Boric acid (H_3BO_3), 30–45 g/L** — a pH buffer that keeps conditions stable at the part’s surface and prevents “burning.”
- **The organic brightener system** — Class I carriers, Class II brighteners, and wetting agent that create the mirror finish and prevent gas pits.

It runs at **pH 3.8–4.2** (mildly acidic), at **130–150°F**, with current densities of **20–60 ASF** (amps per square foot) for rack work.

> ⚠ **HEADS-UP**
>
> The nickel bath is **hot, acidic, and contains nickel — a known skin sensitizer and an inhalation carcinogen.** This is the tank you’ll spend the most time around. Nickel salts cause **allergic contact dermatitis** (“nickel itch”) — once your body becomes sensitized, it is *permanent*. Air agitation throws up a fine **nickel-containing mist** you must not breathe — local exhaust ventilation and mist suppression are mandatory. The OSHA PEL for soluble nickel is **1.0 mg/m³** (as Ni); the stricter ACGIH TLV for soluble nickel is **0.1 mg/m³**, and that lower number is the one to work to on this soluble bath. Always wear gloves; never let solution touch your skin; never breathe the mist. The rectifier runs **high amperage** — lock out before any maintenance.

STAGE 6 — RINSE (POST-PLATE): PROTECT WHAT COMES NEXT (AND RECOVER YOUR NICKEL)

Your part now wears its bright nickel coat, but it’s dripping with nickel bath. If that drag-out goes straight into the chrome tank (Stage 7), it contaminates the expensive chrome chemistry and shortens its life — and it’s literally your nickel going down the drain. So we rinse it off, often with a still **drag-out recovery rinse** first (a stagnant tank that captures the nickel so it can periodically be returned to the bath).

> ⚠ **HEADS-UP**
>
> This rinse is **the decision point for hydrogen-embrittlement baking.** For hardened steel (>40 HRC), parts may need to go from this rinse straight to the **oven — not to the chrome tank** — to be baked within 4 hours per ASTM B850. Skipping or delaying a required bake can lead to parts that crack in service. Always verify the bake requirement and timing on the customer spec before deciding where the part goes next.

STAGE 7 — POST TREATMENT (CHROME / SEAL / LACQUER): LOCK IN THE FINISH

Bright nickel is beautiful, but left bare it slowly **tarnishes and hazes within weeks.** So we add a topcoat — most commonly **decorative chrome** (thin, hard, blue-white, hexavalent Cr^{6+} or the greener trivalent Cr^{3+}), or a **clear lacquer / chromate seal** for parts that don't get chrome.

⚠ **HEADS-UP**

If your post-treatment is **hexavalent chrome (Cr^{6+})**, treat it with the utmost caution: it is a **confirmed human carcinogen** (IARC Group 1), and the OSHA exposure limit is a tiny **5 µg/m³** — *micrograms*, a thousand times smaller than the milligram limits elsewhere. Hex chrome work requires the highest PPE on the line (often supplied-air or PAPR respirators), mandatory mist suppressant, and medical surveillance under OSHA 29 CFR 1910.1026. Trivalent chrome is far less toxic but still demands standard chemical PPE and ventilation. Do not work a chrome tank until specifically trained.

STAGE 8 — FINAL RINSE / DRY: SEND IT OUT SPOTLESS

Last step: a clean **DI (deionized) water** rinse to remove the last traces of chemistry, then a gentle warm-air dry. The goal is a flawless, **spot-free** finish — because after all that work, a single water spot or fingerprint can reject the part.

💡 **TIP**

Handle finished parts with **clean, dry, lint-free cotton or nitrile gloves only.** On fresh chrome, fingerprints can become permanent. The finish you spent eight stages creating can be ruined in the final ten seconds by a bare thumb. Treat finished parts like glass.

★ **REMEMBER**

The whole eight-stage sequence is one continuous logic chain: *clean it so nickel can stick → rinse so the cleaner doesn't ruin the acid → activate so the metal is bare and ready → rinse so the acid doesn't poison the precious nickel → plate the bright nickel → rinse so nickel doesn't waste into chrome → topcoat to protect the nickel → final rinse and dry so it ships flawless.* Every step earns its place. Skip one, and the chemistry of the next step tells on you.

CHAPTER 5

PPE — YOUR DAILY ARMOR

THE GEAR THAT STANDS BETWEEN YOU AND CHEMICALS THAT BURN, MIST THAT HARMS, SPLASHES THAT BLIND

Before you touch a single tank, you get dressed for the job. PPE is **Personal Protective Equipment**. On a plating line, PPE is not optional, not "for the new guys," and not something you skip "just this once." It's your daily armor. Put it on every time, all the way.

THE STANDARD BRIGHT NICKEL LINE PPE

- **Splash goggles** — sealed chemical goggles, *not* regular safety glasses. Glasses leave gaps; a splash finds the gaps. Goggles seal to your face.
- **Face shield** — worn *over* goggles when there's any chance of splash (transferring parts, making additions, working over a hot tank). The face shield protects your face; the goggles protect your eyes. You need both.
- **Chemical-resistant gloves** — **elbow-length nitrile** is the workhorse. Long enough that solution can't run down inside when you reach into a tank.
- **Chemical-resistant apron** — covers your torso and legs from splashes and drips.
- **Chemical-resistant boots** — closed, non-slip, acid/alkali-resistant. Floors near tanks get wet and slippery.
- **Respiratory protection when needed** — a **P100 (or supplied-air) respirator** if ventilation is inadequate around nickel mist, acid fumes, or chrome. For hexavalent chrome, a **PAPR or supplied-air respirator** with proper cartridges is standard.

★ **REMEMBER**

Goggles AND face shield. People remember one and skip the other. Goggles seal your eyes against fine splash and mist; the face shield protects the rest of your face from bigger splashes. They do different jobs. Wear both whenever you work over or near a tank.

⚠ **HEADS-UP**

PPE only protects you if it's in good shape. **Inspect your gloves for pinholes and cracks before every use** — a glove with a hidden hole is worse than no glove, because it traps chemical against your skin. Replace damaged PPE immediately. Take it *off* properly: peel gloves inside-out, wash up before breaks, and never wear contaminated PPE into break rooms.

💡 **TIP**

Match the glove and respirator to the *specific* chemical — check the SDS for the recommended glove material and cartridge type. Nitrile is great for most of the line, but some specialty chemicals attack certain materials. When in doubt, ask your supervisor or check the SDS glove-compatibility chart before you reach in.

THE "NO PERSONAL HABITS" RULE

⚠ **HEADS-UP**

No eating, drinking, chewing gum, smoking, vaping, or applying cosmetics anywhere on the plating line — ever. All of these put your hand to your mouth, and your hands carry chemical residue you can't see. Trace nickel, chrome, or acid ingested over time is a genuine health hazard. Eat and drink only in designated clean areas, and wash your hands and face thoroughly first. This single habit prevents a surprising amount of long-term harm.

CHAPTER 6

EMERGENCY & FIRST-RESPONSE

WHEN SECONDS COUNT — LEARN THIS COLD, BEFORE YOU EVER NEED IT

Even in a well-run shop, accidents happen. The difference between a scare and a serious injury is usually *how fast you respond in the first few seconds.*

★ **REMEMBER**

On your very first day, walk the line and locate every **emergency shower** and **eyewash station.** They must be reachable within about **10 seconds** of every chemical station. You should be able to get to one *with your eyes closed*, because if you take a chemical splash to the face, you *will* have your eyes closed. Know the path by feel.

Also locate: the **SDS binder or station** (Safety Data Sheets for every chemical), the **spill kit**, the **fire extinguisher**, the **first aid kit**, the **emergency stop / rectifier shutoff**, and the **nearest phone or alarm.**

• FIRST RESPONSE BY HAZARD

SITUATION	WHAT TO DO — IMMEDIATELY
Chemical on skin (acid, alkali, or nickel)	Get to the emergency shower **immediately**; flush with copious water for **at least 15 minutes**. Remove contaminated clothing *while* flushing. Don't neutralize a burn with another chemical — just flush. Get medical attention; bring the SDS.
Chemical in eyes	Go to the eyewash **instantly**; flush **at least 15 minutes**, holding eyelids open and rolling your eyes. Have someone get help while you keep flushing — *do not* leave the eyewash. Seek medical attention; bring the SDS.
Breathing in fumes/mist (acid, nickel, chrome)	Move the person to fresh air immediately. If breathing is difficult, call emergency medical help. Report it — ventilation may have failed and needs fixing before anyone returns.
Electrical contact / shock	**Do not touch the person if they may still be in contact with current** — you'll become a second victim. Hit the emergency stop / shut off the rectifier first. Call for emergency help.
Chemical spill	Alert others and keep people away. Only clean it if trained and within your shop's "small spill" limits — otherwise evacuate and call trained spill response. Never let spills reach floor drains.

⚠ **HEADS-UP**

Hydrofluoric acid (HF) — used in some zinc die-cast activators — is a special, deadly case. HF burns may not hurt much at first but can cause severe deep-tissue damage and dangerous changes to your body's calcium that can stop your heart. If HF is used in your shop, **calcium gluconate gel must be on hand**, you must be trained on its use, and any HF exposure is a *call-911-immediately* event even if it looks minor. Never treat HF like an ordinary acid.

💡 **TIP**

When you report an incident — and you should report *every* incident, even a "minor" splash — grab the **SDS** for the chemical involved and have it ready for the medical responder. The SDS has the specific first-aid and treatment information doctors need, and it saves critical time.

★ **REMEMBER**

Flush first, ask questions later. For skin and eye chemical exposure, the single most effective thing you can do is get water on it *fast* and keep it there for a full 15 minutes. Speed beats everything. The shower and eyewash are your best friends on this line.

CHAPTER 7

THE SAFETY PHILOSOPHY OF A NICKEL LINE

THE MINDSET THAT KEEPS YOU SAFE ACROSS A WHOLE CAREER

SAFETY ISN'T A STEP — IT'S THE WHOLE JOB

You may have noticed this manual sprinkles HEADS-UP boxes into *every* section. That's deliberate. In some jobs, safety is a separate task you "do" and move on. **In plating, safety is woven into every single thing you do**, because nearly every tank can hurt you in a different way. There's no "safe part" of the line where you can relax your guard.

★ **REMEMBER**

There is no neutral step on a nickel line — for the chemistry *or* for your safety. Just as every process stage either protects or poisons the next, every task either keeps you safe or exposes you to a hazard. Treat every tank as dangerous until proven otherwise, every time.

KNOW YOUR FOUR BIG HAZARD FAMILIES

1. **Hot alkalis (the cleaners)** — Stage 1. Hot (140–180°F), high-pH (11–13.5) solutions causing chemical burns and eye injury. Plus electrocleaning generates flammable gases.
2. **Strong acids (activation dip and chrome catalysts)** — Stages 3 & 7. HCl and H_2SO_4 cause severe burns, give off fumes, and release flammable hydrogen. HF (if present) is in a lethal category of its own.
3. **Nickel (the plating bath)** — Stage 5. A **skin sensitizer** causing permanent allergic dermatitis ("nickel itch"), and a **carcinogen by inhalation** of its mist (GHS Category 1A; OSHA PEL 1.0 mg/m³ soluble nickel as Ni, stricter ACGIH TLV 0.1 mg/m³). Hot and acidic on top. A *chronic, cumulative* hazard — the damage builds quietly, so complacency is the real enemy.
4. **Hexavalent chromium (chrome post-treatment)** — Stage 7. A **confirmed human carcinogen** (IARC Group 1) with a microscopic exposure limit (5 µg/m³). The most tightly regulated chemical on the line.

⚠ **HEADS-UP**

Across all eight stages: **emergency shower and eyewash within 10 seconds of every station; SDSs accessible at all times; no eating, drinking, or smoking on the line; full PPE every time; lock out / tag out (LOTO) the rectifier before any tank maintenance.** These rules are not bureaucracy — every one is written in someone's past injury. Honor them.

TWO KINDS OF HARM: SUDDEN AND SLOW

- **Acute (sudden) hazards** — a splash burn, an acid fume lungful, an electric shock. They get your attention instantly; your fast response (flush, fresh air, LOTO) protects you.
- **Chronic (slow) hazards** — nickel sensitization, repeated low-level mist exposure, long-term chrome exposure. No pain today, which is exactly why people get careless. The harm accumulates silently and can be *permanent.*

★ **REMEMBER**

The hazard that doesn't hurt today is often the one that hurts you for life. Nickel sensitization and carcinogen exposure don't announce themselves with pain — so you can't rely on "it doesn't bother me" as your safety gauge. Wear the gloves and the respirator *every* time, even when nothing feels wrong.

LOCK OUT, TAG OUT (LOTO)

⚠ **HEADS-UP**

LOTO means physically locking a power source in the "off" position and tagging it with your name before you work on equipment, so no one can energize it while your hands are in harm's way. The rectifier pushes huge current into conductive, liquid-filled tanks. **Never reach into a tank, change anodes, or service plating equipment until the power is locked out and tagged out — by you, with your own lock.** "I'll just be a second" is exactly how people get hurt. No exceptions, ever.

💡 **TIP**

The best safety habit of all is humility. Nobody — not even a 20-year veteran — knows everything about every chemical and situation. If something looks wrong, smells wrong, or you're simply not sure, **stop and ask your supervisor.** A two-minute question is always cheaper than an accident. The operators you'll admire most are the ones who *never* stopped checking.

★ **REMEMBER**

You've now got the whole foundation: what plating is, what makes bright nickel special, how the eight-stage line works as one connected system, why the order is locked by chemistry, what to wear, what to do in an emergency, and the safety mindset that ties it all together. Everything in the stage chapters builds on these fundamentals. Turn the page knowing the *why* — and the *how* will make a lot more sense.

PART TWO — THE PREP LINE (STATIONS 01–04)

STATION 01 OF 08

SOAK CLEAN / ELECTROCLEAN

THE FIRST TANK — AND THE MOST UNDERRATED ONE

★ **REMEMBER — THE PREP-LINE PROMISE**

By the time a part reaches the nickel tank, its fate is mostly decided. Bright nickel is a tattletale — it magnifies every flaw into a finish defect you can see across the room. The prep line (clean, rinse, activate, rinse) isn't the "boring stuff before the real plating." It *is* the real plating.

Picture a brand-new part arriving at your line. It looks clean — maybe even shiny. But under a microscope that "clean" surface is a greasy battlefield: stamping oils, drawing lubricants, buffing compound, fingerprints, shop dust, and a thin skin of metal oxide. None of that is visible to your eye; all of it is visible to bright nickel. Your job at Station 01 is to strip every last molecule of it off so the surface is *chemically naked* — ready to bond.

☀ **FUN FACT**

The word "clean" in plating doesn't mean "looks clean" — it means "water-break free." A surface can sparkle and still fail. We measure cleanliness with water, not with our eyes.

WHAT YOU'RE DOING & WHY IT MATTERS

Soak cleaning is exactly what it sounds like: the part soaks in a tank of hot **alkaline cleaner** (high-pH, like strong dish soap with muscles). Its **surfactants** (soap-like molecules) surround oil droplets and lift them off (*emulsification*). Its **builders** (alkaline salts) break down greasy soils chemically. Heat speeds it all up — that's why the tank runs hot.

Electrocleaning comes next and is the step new operators most often want to skip. *Don't.* We hang the part on a cathode rod, dunk it in cleaner, and run DC through the solution. The current splits water into gas bubbles at the part surface, which do two things: **mechanical scrubbing** (millions of tiny bubbles tearing away the last invisible film) and **electrochemical lift** (the surface charge repels oily contaminants). Run the part **cathodic first** (negative, generates hydrogen — twice the bubble volume, twice the scrubbing), then flip to **anodic** for the last 15–30 seconds (positive, generates oxygen).

⚙ **TECHNICAL STUFF**

Why finish anodic? Cathodic cleaning is the better scrubber, but it can plate microscopic metallic "smut" back onto your part from impurities in the cleaner, and it pumps hydrogen *into* the metal. The final anodic burst strips that smut back off and reverses the gas direction — the difference between a surface that looks clean and one that actually bonds. This sequence is called *reverse-current* or *periodic-reverse* cleaning.

★ **REMEMBER**

Soak removes the *bulk* soil. Electroclean removes the *microscopic film* that causes pitting and skip plating. They are two different jobs. Bright nickel demands both.

THE NUMBER THAT RULES THIS STATION: ZERO SECONDS

The single most important spec at Station 01 isn't a temperature or a time — it's the **water-break test**, target **0 seconds.** Pull a cleaned part, hold it vertical ~30 seconds, and watch the water film. On a truly clean surface, water spreads into a **continuous, unbroken sheet** — zero seconds of beading. If even one spot **beads up** like rain on a waxed car, oil is still there. That bead is invisible contamination made visible.

```
FAIL - water beads up          PASS - water sheets evenly
+--------------+               +--------------+
|  o   o   o   |  X            |##############|  OK
|    o    o    |               |##############|
+--------------+               +--------------+
Oil remains -> RE-CLEAN        Surface clean -> proceed
```

💡 **TIP**

The water-break test is free, takes 30 seconds, and catches problems before they cost you a whole rack of scrap. Test *every rack* — not just the first one of the day. Cleaners get tired as the shift goes on.

OPERATOR ESSENTIALS

MEMORIZE THE RANGES

PARAMETER	RANGE	WHAT TO WATCH
Soak clean temperature	140–180°F (60–82°C)	**Never exceed 200°F** — damages some cleaners
Soak clean time	3–10 min	Heavy draw oil: 8–10 min · Light stamping oil: 3–5 min
Soak concentration	4–8 oz/gal (30–60 g/L)	Titrate **at least weekly**
Electroclean voltage	4–8 V	Current density typically 30–80 ASF
Electroclean time	1–3 min	**Never exceed 3 min cathodic on high-strength steel**
Electroclean polarity	Cathodic → Anodic	Reverse the **last 15–30 sec** to strip smut
Electroclean temperature	140–180°F (60–82°C)	Same range as soak
pH	11.0–13.5	Depends on your cleaner formulation
Water-break test	Pass = continuous sheet (0 s)	**Mandatory before proceeding**

★ **REMEMBER**

“ASF” means **amps per square foot** — the amount of plating/cleaning current spread over the surface area of your parts. You’ll see ASF everywhere on this line. Higher ASF = more intense action.

KNOW YOUR SUBSTRATE (THE METAL CHANGES THE RECIPE)

SUBSTRATE	SOAK CLEANER	ELECTROCLEAN	WATCH FOR
Mild steel	Standard alkaline, 140–180°F	Cathodic → anodic finish	Rust flash if transfer is slow
High-strength steel	Standard alkaline, 140–160°F	**Limit cathodic to 60 sec max**	Hydrogen embrittlement from excess cathodic
Brass / copper	Mild alkaline, 120–140°F	Anodic only (short)	Tarnish; dezincification of brass
Zinc die cast	Mild **non-etch**, 120–140°F	Anodic only, **<30 sec**	Etching; hydrogen blistering of the casting
Stainless steel	Standard alkaline, 160–180°F	Anodic preferred	Passive film — may need a Wood’s nickel strike

⚙ **TECHNICAL STUFF**

“Dezincification” is when aggressive chemistry selectively dissolves zinc out of brass, leaving a weak, spongy copper-rich surface. “Hydrogen embrittlement” is when atomic hydrogen soaks into high-strength steel and makes it brittle enough to crack later under load — sometimes days after the part ships. Both are silent killers. This is *why* we limit cathodic time on those metals.

STEP-BY-STEP PROCEDURE

1. **Verify the tank is ready.** Temperature 140–180°F, cleaner titrated this week reading 4–8 oz/gal. If you don’t know how to titrate, ask *before* running parts.
2. **Load the rack** so solution reaches every surface — no trapped air pockets, no parts shielding each other.
3. **Soak** 3–10 minutes based on how oily the parts are. Light agitation helps.
4. **Drain** over the soak tank a few seconds so you don’t carry a flood of cleaner forward.
5. **Electroclean.** Run **cathodic** at 4–8 V for most of the cycle, then **reverse to anodic for the final 15–30 seconds.** Total 1–3 minutes. Respect substrate limits.
6. **Water-break test.** Confirm a continuous sheet. **Bead = stop and re-clean.** Don’t rationalize a borderline result.
7. **Move on — promptly** to Station 02. Do **not** let cleaned parts air-dry; a dry, freshly cleaned surface flash-oxidizes fast.

• TEACH-BACK CHECKLIST

A trainee is signed off on Station 01 only when they can, unprompted:

- [ ] State the soak temperature range and the hard ceiling (**140–180°F; never over 200°F**).
- [ ] Explain *why* both soak **and** electroclean are required (bulk soil vs. microscopic film).
- [ ] Correctly perform a water-break test and read the result honestly.
- [ ] Demonstrate the **cathodic-then-anodic** sequence and explain why we finish anodic.
- [ ] Identify which substrates need limited cathodic time and explain the embrittlement/blistering risk.
- [ ] Verbally walk through what they'd do if the water-break test fails.

• TOP MISTAKES NEW OPERATORS MAKE

1. **Skipping the electroclean** because the soak "looked good." The #1 cause of downstream pitting and skip plating.
2. **Trusting their eyes instead of the water-break test.** Shiny ≠ clean.
3. **Touching cleaned parts with bare hands.** A fingerprint is oil. You just re-contaminated the part. Clean gloves, always.
4. **Letting parts air-dry** between cleaning and rinsing.
5. **Running tired cleaner.** If concentration or temperature drifts low, cleaning quietly fails. Titrate on schedule and skim floating oil.
6. **Over-cathodic-cleaning high-strength steel or die cast**, inviting embrittlement or blistering.

⚠ **HEADS-UP — SAFETY, STATION 01**

Hot alkaline solution causes chemical burns on contact — and alkaline burns are sneaky because they don't always hurt immediately while they keep damaging tissue. Full PPE: splash goggles, face shield, **elbow-length** chemical gloves, apron, chemical-resistant boots.

Electrocleaning generates hydrogen and oxygen gas — a flammable mix. Ventilation required and **no open flames near the tank, ever.** The electrocleaner is **DC current running through liquid — LOTO before reaching into the tank.**

First aid: **skin contact → flush 15 minutes; eye contact → eyewash 15 minutes** and get medical help. Know where your shower and eyewash are *before* you need them.

SIGN-OFF — STATION 01

Trainee: ____________________ Trainer: ____________________ Date: ____________

"I can run a soak/electroclean cycle, perform and honestly interpret a water-break test, and I understand the burn, gas, and electrical hazards at this station."

STATION 02 OF 08

RINSE (PRE-ACTIVATION)

THE FIREWALL — WHY A "JUST RINSING" STEP DESERVES ITS OWN CHAPTER

It's tempting to think of rinse tanks as plumbing — water in, water out, nobody cares. That mindset will wreck your line. Station 02 is a **firewall**: it stands between your hot alkaline cleaner and the acid tank that comes next, and ultimately the nickel bath beyond that.

When a part leaves the cleaner, it carries a thin film of cleaner — we call this **drag-out** (chemistry literally dragged out of the tank on the part). If you let that alkaline drag-out ride into the acid tank, two bad things happen at once: the cleaner **neutralizes your acid** (acid + base cancel out), so you burn through expensive acid for nothing; and the cleaner's hidden ingredients — **surfactants, silicates, and sodium** — get a free ride deeper down the line, eventually into the nickel bath.

> ★ **REMEMBER**
> Drag cleaner into the acid and you **waste acid.** Drag cleaner into the nickel and you **poison it.** This rinse is the only thing standing in the way.

• WHAT YOU'RE DOING & WHY IT MATTERS

Rinsing is dilution. You're swapping the concentrated cleaner film for clean water, over and over, until what's left is negligible. We measure success with **conductivity** — a meter that reads how many dissolved ions are in the water. Pure water barely conducts; dirty rinse water conducts a lot. So **low conductivity = clean rinse.** The unit is **µS/cm** (microsiemens per centimeter — a contamination score; lower is cleaner).

> ★ **REMEMBER**
> **Keep the pre-activation rinse below 500 µS/cm — aim for under 300.**

Why do those hidden cleaner ingredients matter so much downstream? Each one is a specific saboteur:

- **Surfactants** (soaps) cause foaming and gas **pitting** in the nickel bath.
- **Silicates** form invisible, insoluble films that **block adhesion** — the nickel literally won't stick.
- **Sodium** accumulates in the nickel bath **permanently** and has no business being there.

• THE SMART WAY TO RINSE: COUNTER-FLOW

You *could* just blast fresh water through a single tank and dump it. It works, but wastes enormous water. The professional approach is a **counter-flow (counter-current) rinse**: the parts move forward; the water flows backward. Fresh water enters the *last* (cleanest) tank, then overflows backward into the dirtier tank. So your part gets progressively cleaner rinses, and the cleanest water it ever sees is the last thing to touch it before the acid tank.

```
FROM        +-------------+  PARTS ->  +-------------+        TO
CLEANER ->  |  RINSE 1    | ---------> |  RINSE 2    | ---->  ACID
            |  (dirty)    |            |  (clean)    |
            | <500 uS/cm  |            | <200 uS/cm  |
            +-------------+            +-------------+
                 ^   overflow <----------    ^
                 +------ <- FRESH WATER FLOWS BACKWARD <- -----+
                                                      FRESH IN
```

> ☀ **FUN FACT**
> The math is striking. A single rinse tank gives roughly a **30:1** dilution. A two-stage counter-flow rinse gives an *effective* **~900:1** — and uses **70–85% less water.** Same cleanliness, a fraction of the water bill. Engineering at its finest.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient (60–85°F / 16–29°C)	No heating required
Time	30–60 sec with agitation	Longer for blind holes, tubing, recesses
Water source	Municipal or DI	DI preferred for nickel lines
Conductivity	<500 µS/cm (target <300)	**Monitor daily**
Flow rate	2–4 gal/min	Continuous overflow or counter-current
Tank turnover	4–6 per hour	Prevents stagnation
pH	7–10	Elevated pH means cleaner is dragging in

TIP

“DI” is **deionized water** — water with the dissolved minerals stripped out. Municipal tap water carries calcium, chloride, and other junk you don’t want creeping toward the nickel bath. On a bright nickel line, lean toward DI wherever you can.

TIP

Watch the rinse **pH** as an early-warning gauge. If it climbs above ~10, your cleaner drag-out is winning and your firewall is leaking. Tighten drain time or add agitation before it becomes a downstream problem.

• STEP-BY-STEP PROCEDURE

1. **Drain over the cleaner first.** Let the part drain over the cleaning tank a few seconds. Less drag-out = easier rinsing and recovered chemistry.
2. **Immerse with agitation** 30–60 seconds, moving the rack. Agitation flushes recesses, blind holes, and threaded areas; still water just sits there.
3. **Give complex parts extra time.** Tubing, deep recesses, and blind holes trap cleaner. Add dwell and work the rack up and down.
4. **Check conductivity.** Confirm under 500 µS/cm (ideally under 300). If creeping up, increase fresh-water flow.
5. **Drain and advance promptly** to the acid tank. Don’t let the part dry.

• TEACH-BACK CHECKLIST

- [ ] Explain what “drag-out” is and the two distinct downstream harms (waste acid; poison nickel).
- [ ] State the conductivity target (**<500 µS/cm, aim <300**) and where to read it.
- [ ] Name the three hidden cleaner ingredients and the damage each causes (surfactant→pitting, silicate→adhesion failure, sodium→permanent contamination).
- [ ] Describe how a counter-flow rinse works and why water flows opposite to the parts.
- [ ] Demonstrate proper agitation and extra dwell for a part with blind holes.

• TOP MISTAKES NEW OPERATORS MAKE

1. **Treating the rinse as “just water”** and not monitoring conductivity.
2. **No agitation** — letting parts sit dead-still so recesses never get flushed.
3. **Skipping the drain over the cleaner**, dumping a flood of drag-out into rinse #1.
4. **Letting the rinse go stagnant** — too little fresh-water flow; conductivity quietly climbs all shift.
5. **Ignoring rising rinse pH**, the earliest sign the firewall is failing.

⚠ HEADS-UP — SAFETY, STATION 02

This is the gentlest station on the prep line, but “gentlest” is not “harmless.” The rinse water holds **residual alkaline cleaner** dragged in from Station 01, so it can still irritate skin and eyes. Wear **goggles, chemical gloves, and non-slip footwear.**

The real everyday hazard here is **slips and falls** — rinse stations make floors wet. Keep the area squeegeed, wear non-slip boots, and walk, don’t rush. No special ventilation required at this stage.

SIGN-OFF — STATION 02

Trainee: ______________________ Trainer: ______________________ Date: ______________

“I understand why this rinse is the firewall, I can read and act on conductivity, and I know the slip and residual-cleaner hazards.”

STATION 03 OF 08

ACID ACTIVATION

THE ACID DIP — THE MOST COUNTERINTUITIVE STEP ON THE LINE

By now your part is spotlessly clean. So why are we about to dunk it in acid? Because “clean” and “ready to plate” aren’t the same thing. The instant bare metal is exposed to air and water — even during cleaning and rinsing — it grows a microscopically thin layer of **oxide** (metal combined with oxygen, the same family of reaction as rust, just invisibly thin). That oxide is a barrier. Nickel cannot bond to oxide; it can only bond to live, bare metal. **Activation** dissolves that oxide skin and exposes a chemically “hungry” surface that nickel will grab onto.

★ **REMEMBER**

The acid dip is not cleaning — the cleaning is already done. It’s **de-oxidizing.** You’re removing an invisible film so the nickel has bare metal to bond to. *Oxide is invisible. Adhesion failure is not.*

• WHAT YOU’RE DOING & WHY IT MATTERS

The acid chemically dissolves the surface oxide and leaves an electrochemically **active** surface — hence the name. Get the dose right and nickel adheres beautifully. Get it wrong either direction and you pay:

- **Under-dip** (too short/weak): oxide survives → nickel won’t stick → **adhesion failure** (peeling, blistering).
- **Over-dip** (too long/strong/hot): you stop dissolving oxide and start dissolving *the metal itself* → **etching and pitting**, plus excessive **hydrogen loading.**

★ **REMEMBER**

Activation is a Goldilocks step. The window is **15–60 seconds** on steel — not “a little is good so more is better.” More is worse. **Steel dip time: 15–60 s** (light oxide 15–30 s; heavy scale 30–60 s). Copper alloys and zinc die cast get **much** less — 5–15 s — because acid attacks them fast.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient–110°F (Amb–43°C)	Ambient typical for mild activation
Time — steel	15–60 sec	Light oxide 15–30 s · heavy scale 30–60 s
Time — brass/copper	5–15 sec	Short — acid attacks copper alloys quickly
Time — zinc die cast	5–15 sec	Minimal; fluoride activator preferred
Acid Option A	HCl 10–50% v/v	Most common; **fumes at room temp — ventilation required**
Acid Option B	H_2SO_4 5–20% v/v	Less fuming; introduces **no chloride**
Acid Option C (die cast)	10–20% H_2SO_4 + 1–2% HF	Fluoride activates zinc die cast without over-attacking base metal
Iron content	Dump when >50 g/L Fe	Spent acid loses effectiveness and drags metal into nickel
Agitation	Manual rack movement only	**No air agitation** — it creates an acid-mist hazard

TECHNICAL STUFF

"% v/v" means **percent by volume** — e.g., 10% v/v HCl is 10 parts concentrated acid in 100 parts total solution. **Always add acid to water, never water to acid** — adding water to concentrated acid can boil and spit acid back at you. The memory aid: *"Do as you oughta — add acid to water."* HCl works fast but fumes and adds **chloride** that drifts toward the nickel bath; H_2SO_4 fumes less and adds **sulfate**, which the nickel bath already contains — so sulfate is the friendlier guest downstream.

• CHOOSE THE RIGHT ACID FOR THE METAL

SUBSTRATE	RECOMMENDED ACID	TIME	KEY PRECAUTION
Mild steel	HCl 10–50% v/v	15–60 s	Use **inhibited** acid to limit hydrogen absorption
High-strength steel	HCl 10–20% v/v (inhibited)	15–30 s	**Minimum** effective time — bake after plating
Brass	H_2SO_4 5–10% v/v	5–15 s	**Avoid HCl** — dezincification risk
Zinc die cast	Fluoride-based (per TDS)	5–15 s	**No straight HCl/H_2SO_4** — use the proprietary activator
Copper	H_2SO_4 5–10% v/v	5–10 s	Brief dip only — copper dissolves quickly

TIP

"Inhibited" acid contains an additive (corrosion inhibitor) that lets the acid keep eating oxide while slowing its attack on the base metal and cutting hydrogen absorption. On high-strength steel, inhibited acid isn't optional — it's how you protect the part from cracking later.

TECHNICAL STUFF — THE HYDROGEN-EMBRITTLEMENT WARNING

Acid activation is a major source of hydrogen soaking into steel. For high-strength steel (**>40 HRC / >180 ksi UTS** — ultimate tensile strength), use the **minimum** effective acid concentration and time, always use **inhibited** acid, and **bake the parts after plating per ASTM B850 — 375 ±25°F, within 4 hours** of the last plating step. This is a safety-of-the-finished-part issue: an embrittled bolt can fail catastrophically in service days after it ships.

• STEP-BY-STEP PROCEDURE

1. **Confirm the part came straight from a good rinse** and isn't dried out. Oxide grows on a dry surface.
2. **Verify the acid type and strength** match the substrate. Confirm the tank isn't spent (iron over 50 g/L = dump it).
3. **Immerse for the correct time** — short for copper alloys and die cast, longer for scaled steel. Watch the clock; don't eyeball it.
4. **Agitate by hand** — gentle rack movement only. **Never** turn on air agitation in an acid dip; it aerosolizes acid into the air you breathe.
5. **Drain over the acid tank** briefly to recover acid and reduce drag-out.
6. **Advance immediately** to Station 04. The freshly activated surface is reactive and will re-oxidize if it sits.

• TEACH-BACK CHECKLIST

- [ ] Explain what activation does (dissolves oxide; exposes bare, reactive metal) and why clean ≠ ready-to-plate.
- [ ] State the steel dip time (**15–60 s**) and why both under-dip and over-dip are bad.
- [ ] Choose the correct acid for steel, brass, and zinc die cast and justify the choice.
- [ ] Explain why **no air agitation** is allowed in the acid tank.
- [ ] State the hydrogen-embrittlement rule for high-strength steel and the post-plate bake (**375 ±25°F within 4 hours, ASTM B850**).
- [ ] Recite the "add acid to water" rule.

• TOP MISTAKES NEW OPERATORS MAKE

1. **Over-dipping** because "more must be cleaner" — etches the part and loads it with hydrogen.
2. **Using HCl on brass**, dezincifying the surface and creating copper smut.
3. **Turning on air agitation** in the acid tank — a serious inhalation hazard.
4. **Running spent, iron-loaded acid** that no longer activates and feeds iron toward the nickel bath.
5. **Forgetting the bake** on high-strength steel, risking embrittlement cracking in service.
6. **Letting activated parts sit** and re-oxidize before plating.

⚠ **HEADS-UP — SAFETY, STATION 03 (THE MOST HAZARDOUS TANK ON THE PREP LINE)**

Strong mineral acid — causes severe burns and permanent eye damage. HCl fumes at room temperature — you can be exposed just standing over the tank. **Hydrogen gas from pickling is flammable.** Ventilation is mandatory; no open flames.

If your shop activates zinc die cast with **HF (hydrofluoric acid): treat it as extraordinarily dangerous.** HF penetrates skin and attacks bone and the bloodstream; a seemingly minor splash can be fatal. **Calcium gluconate gel must be on hand and within reach before anyone runs HF**, and everyone must know the HF-specific first-aid protocol.

PPE: **goggles + full face shield**, acid-resistant gloves, apron, chemical boots, and a **respirator if ventilation is inadequate.** Remember: **always add acid to water, never the reverse.** Know your eyewash and shower locations cold.

SIGN-OFF — STATION 03

Trainee: ______________________ Trainer: ______________________ Date: ____________

"I can select and run the correct activation for the substrate, I understand the burn, fume, hydrogen, and (if applicable) HF hazards, and I know the embrittlement-bake rule for hardened steel."

STATION 04 OF 08

RINSE (PRE-PLATE)

THE GUARD RINSE — THE MOST VALUABLE TANK ON THE LINE, AND IT'S "JUST" A RINSE

Welcome to the gatekeeper. Station 04 is the **last barrier before the bright nickel bath**, and that makes it the single most important contamination-control point on the whole line.

★ **REMEMBER**

The bright nickel bath runs hot (**130–150°F**) and **never gets dumped.** A well-maintained Watts bath can run for **years — even decades.** Every milligram of acid, dissolved iron, copper, or zinc you let slip through this rinse **accumulates permanently.** There is no "flush it out next week." It's forever. That's why we call it the **guard rinse.**

• WHAT YOU'RE DOING & WHY IT MATTERS

Your part just came out of the acid tank carrying acidic drag-out loaded with dissolved metals (especially **iron** from the pickle). The guard rinse dilutes all of that away before the part enters the nickel. Same conductivity principle as Station 02, but the target is **much tighter** because the stakes are permanent:

★ **REMEMBER**

Pre-plate rinse: keep conductivity below 200 µS/cm — aim for under 100. This is the tightest rinse spec on the prep line, by design.

What specifically are we keeping out, and why does each one matter?

CONTAMINANT	SOURCE	EFFECT ON THE NICKEL BATH	PREVENTION
Iron (Fe)	Spent pickle	Haze, reduced brightness above **25 ppm**	Refresh pickle; double rinse
Chloride (Cl^-)	HCl activation	Excess raises anode corrosion and pitting	Switch to H_2SO_4; tighter rinse
Fluoride (F^-)	Die cast activator	Attacks the boric-acid buffer; destabilizes pH	Dedicated rinse after fluoride dip
Sulfate (SO_4)	H_2SO_4 activation	Minor — already present in the Watts bath	Normal rinse sufficient

⚙ **TECHNICAL STUFF — WHY IRON IS THE NIGHTMARE GUEST**

Iron dissolves readily in the hot, acidic Watts bath, and above **~25 ppm** it causes haze and kills brightness. The cruel part: iron **cannot be easily removed** once it's in. (Purification means raising bath pH to oxidize and precipitate the iron, then filtering — a production-stopping procedure.) Far easier to keep it out. The guard rinse is your primary defense, backed by keeping the pickle fresh.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	`Ambient (60–85°F)`	No heating required
Time	`30–60 sec with agitation`	**Double rinse strongly recommended**
Water source	`DI strongly preferred`	Municipal adds calcium, chloride, contaminants
Conductivity	`<200 µS/cm (target <100)`	Critical for nickel bath longevity
pH	`4.0–8.0`	**Below 4 = excessive acid drag-in**
Flow rate	`3–5 gal/min`	Counter-current ideal
Drain time	`15–20 sec minimum`	Prevents diluting the nickel bath

💡 **TIP**

The drain step does double duty: a solid 15–20-second drain over the rinse both reduces acid carry-in *and* keeps water from diluting the precious nickel bath. Two birds, one pause.

THE 30-SECOND CLOCK: GET TO NICKEL FAST

There's a timing trap unique to this station. The part is now clean, activated, and *bare* — which means it's also eager to re-oxidize the moment it dries. If you rinse beautifully and then let the part hang in the air, it can oxidize *before* it reaches the nickel and you've undone Station 03.

★ **REMEMBER**

Transfer from the guard rinse into the nickel bath within 30 seconds. Keep the part wet. No coffee breaks between Station 04 and Station 05.

STEP-BY-STEP PROCEDURE

1. **Drain over the acid tank** before entering the rinse — recovers acid and cuts the contaminant load you're about to rinse off.
2. **Immerse with agitation** 30–60 seconds. **Use a double rinse** if your line has one — the second stage is what gets you under 100 μS/cm.
3. **Check conductivity** and confirm under 200 μS/cm (target <100). Check pH stays at or above 4 — a low reading means acid is breaking through.
4. **Drain 15–20 seconds** over the rinse.
5. **Go straight to the nickel bath — within 30 seconds.** Keep the surface wet the whole way.

TEACH-BACK CHECKLIST

- [ ] Explain why this is "the most valuable tank on the line" (bath never dumped; contamination is permanent).
- [ ] State the conductivity target (**<200 μS/cm, aim <100**) and how/why it differs from Station 02.
- [ ] Name the iron action level (**25 ppm**) and explain why iron is so hard to remove once it's in.
- [ ] Explain the **30-second transfer rule** and what happens if a part dries before plating.
- [ ] Demonstrate proper drain time and (if available) double-rinse technique.

TOP MISTAKES NEW OPERATORS MAKE

1. **Letting conductivity drift** toward the looser Station 02 numbers — this rinse demands tighter control.
2. **Slow transfer to the nickel tank**, letting the activated surface re-oxidize.
3. **Skipping the double rinse** when one is available.
4. **Short-draining**, carrying acid into the nickel and pulling the bath pH down.
5. **Using tap water** where DI is available, quietly feeding chloride and calcium toward the bath.
6. **Ignoring climbing iron or chloride** in the nickel bath instead of fixing the pickle and rinse that feed it.

⚠ **HEADS-UP — SAFETY, STATION 04**

The chemical hazard here is mild — **dilute acid residue** in the rinse water — but still wear **goggles, chemical gloves, and non-slip footwear.** As at every rinse, **wet floors are a slip hazard;** keep the area squeegeed.

The station-specific caution is about **pace, not chemistry:** because parts must reach the nickel tank within 30 seconds, this is a spot where people rush. **Move with purpose, not panic** — a dropped rack or a slip helps no one. Plan your transfer path before you pull the part.

SIGN-OFF — STATION 04

Trainee: ______________________ Trainer: ______________________ Date: ____________

"I understand the guard rinse is the permanent contamination barrier for the nickel bath, I can hold conductivity under 200 μS/cm, and I will transfer parts to nickel within 30 seconds without compromising safety."

★ **REMEMBER — THE PREP-LINE PROMISE**

Four stations, one mission: hand the nickel bath a surface that is **clean, contaminant-free, oxide-free, and still wet.** Clean it (01), rinse the cleaner off (02), strip the oxide (03), rinse the acid off (04), and run to the tank. Do these four right, every rack, every time — and bright nickel will reward you with the mirror finish it's famous for. Cut corners here, and no amount of bath chemistry downstream can save the job.

PART THREE — THE MAIN LINE (STATIONS 05-07)

STATION 05 OF 08 · THE MAIN TANK

BRIGHT NICKEL PLATING

WHERE THE MAGIC FINALLY HAPPENS — THE WATTS BATH

Welcome to the heart of the whole operation. Everything you did in Stations 1 through 4 was preparation. This is the station where your part actually *becomes* a plated part — where a dull, gray, freshly-activated piece of steel walks into a tank and comes out looking like a mirror.

Think of the four stations before this like prepping a wall before painting. You scraped, sanded, wiped, and taped. Nobody hangs that wall in a museum — but if you *skip* the prep, the paint peels and everyone sees it. Station 5 is where you finally pick up the roller — except instead of paint, you're growing solid nickel metal onto the surface, one atom at a time, using electricity.

★ **REMEMBER**

Bright nickel is the most widely used nickel plating process in the world, and it's been around for more than 80 years. The chemistry is rock-solid. That does *not* mean it runs itself. The bath is reliable; the *operator* is the variable. Your attention is what keeps it producing good parts.

• WHAT IS ACTUALLY HAPPENING IN THE TANK?

Electroplating means using electricity to coat one metal with another. We dissolve nickel in a tank of liquid (the **bath** or **solution**), hang your part in it, run current through it, and the dissolved nickel sticks to your part as solid metal. The two characters in this play:

- The **cathode** ("CATH-ode") is your part — connected to the *negative* side. **Memory trick: the cathode is the negative electrode, and it's where the metal builds up.**
- The **anode** ("AN-ode") is the source of fresh nickel — chunks of nickel metal hanging in baskets, connected to the *positive* side. As you plate, the anode slowly dissolves to replace the nickel that leaves the solution.

The power supply that drives all this is the **rectifier** — it takes ordinary wall electricity (which alternates back and forth) and turns it into steady one-direction electricity (**DC**, direct current). Think of the rectifier as the engine of the tank — it sets how fast metal deposits.

⚙ **TECHNICAL STUFF**

The reaction at your part is Ni^{2+} + 2 electrons → Ni^0. A nickel ion floating in solution (positive because it's missing two electrons) grabs two electrons at the surface and becomes a solid, neutral nickel atom locked into the coating. At the anode the reverse happens: solid nickel gives up two electrons and floats off as a fresh ion. The bath is a recycling machine.

☀ **FUN FACT**

The "Watts bath" you're standing in front of is named after Professor Oliver P. Watts, who published the recipe back in 1916. More than a century later, the basic formula has barely changed. When something works *that* well for *that* long, you don't argue with it — you learn to run it right.

• THE BATH RECIPE: WHAT'S IN THE TANK AND WHY

A bright nickel Watts bath is really just five things doing five jobs. Once you know what each ingredient *does*, the names stick.

1. NICKEL SULFATE — $NiSO_4 \cdot 6H_2O$ — THE MAIN METAL SOURCE

The big one. The primary supply of nickel in the tank; the vast majority of the metal that lands on your part comes from it. **Rack:** 240–330 g/L (~32–44 oz/gal) · **Barrel:** 270–330 g/L (kept higher).

> 💡 **TIP**
> You'll hear both "g/L" and "oz/gal" on the floor. Grams per liter is the lab's language; ounces per gallon is old-school American shop language. They describe the same thing. Learn to read both so you're never lost.

2. NICKEL CHLORIDE — $NiCl_2 \cdot 6H_2O$ — THE ANODE'S HELPER

Two quiet but vital jobs. First, it helps the *anodes* dissolve cleanly so they keep feeding fresh nickel — without enough chloride, the anodes "passivate" (grow a film and stop dissolving), starving the bath. Second, it improves **throwing power** — the bath's ability to plate evenly into recessed, low-current areas (inside corners, deep grooves). **Rack:** 30–60 g/L (~4–8 oz/gal) · **Barrel:** 45–60 g/L.

3. BORIC ACID — H_3BO_3 — THE BODYGUARD AGAINST BURNING

This ingredient doesn't end up in your coating at all, yet you cannot run the bath without it. Boric acid is a **buffer** — it resists pH change right at the part's surface. As nickel plates, the chemistry at the surface tries to go more alkaline; if it swings too far you get **burning** (rough, dark, powdery, ruined deposit, especially on edges and high points). Boric acid holds that thin surface layer steady. **Rack:** 30–45 g/L · **Barrel:** 37–45 g/L · **hard floor: never below 30 g/L.**

> ⚠ **HEADS-UP — BORIC ACID**
> Boric acid is a reproductive hazard (GHS Category 1B). Avoid ingestion and airborne dust when adding or handling it, and wear gloves and eye protection.

> ⚠ **HEADS-UP**
> Boric acid only stays dissolved when the bath is warm. If the tank cools below about 100°F — say, over a long weekend with the heater off — the boric acid can **crystallize** out as white crystals on the tank bottom and heaters. When it does, your buffer protection is gone and your next parts will burn. If you find a cold tank with white crystals, do not just turn on the rectifier and plate. Get it back up to temperature, let everything redissolve, confirm the level, and tell your supervisor.

4. THE BRIGHTENER SYSTEM — THREE ORGANIC ADDITIVES

The secret sauce, and where most of your daily attention goes. Without them you'd get plain, dull, gray matte nickel. With them — kept in balance — you get a mirror finish straight from the tank. Full section on the next page.

5. PH — THE ACIDITY OF THE BATH

pH is a number from 0 to 14: below 7 is acidic, above 7 alkaline, 7 neutral. A bright nickel bath lives in a narrow acidic window: **pH 3.8–4.2**, rack and barrel. To bring pH *down* add dilute sulfuric acid (H_2SO_4); to bring it *up* add nickel carbonate ($NiCO_3$) or nickel hydroxide.

> ⚠ **HEADS-UP**
> **Never use sodium hydroxide (caustic soda / lye) to raise the pH of a nickel bath.** It dumps sodium into the bath, where it has no business being and never comes out — a permanent contaminant. Always use nickel-based chemicals to adjust up. This is exactly the kind of adjustment you should never make on your own without sign-off.

THE BRIGHTENER SYSTEM — THREE ADDITIVES, ONE MIRROR

The thing that makes “bright nickel” *bright* is a trio of organic (carbon-based) chemicals dosed in tiny amounts. Think of them as the three legs of a stool — if any one leg is the wrong length, the whole thing wobbles.

COMPONENT	FUNCTION	IF LOW	IF HIGH
Carrier (Class I)	Grain refinement, leveling, sulfur co-deposit	Poor leveling, coarse, dull mid-CD	Excess stress, brittle, peeling
Primary (Class II)	Brightness across the CD range	Dull across all current densities	Brittle, cracking, dark low-CD
Wetting Agent	Pit prevention, lowers surface tension	Gas pitting, especially high CD	Foam, haze, reduced brightness

A quick word on each: the **Carrier** refines the grain (fine, smooth crystals), provides **leveling** (fills tiny scratches so the surface comes out smoother than it went in), and co-deposits a controlled trace of sulfur. The **Primary Brightener** delivers actual brightness and reflectivity across the whole part. The **Wetting Agent** lowers the bath’s surface tension so hydrogen bubbles don’t cling to the part and leave **pits** (pinholes).

TECHNICAL STUFF

Carriers are typically saccharin or sulfonamide compounds (yes, saccharin — the same family as the old artificial sweetener). They deliberately put a small amount of sulfur into the nickel. That sulfur is what makes the coating “active,” and it’s exactly what makes a two-layer “duplex” nickel system protect against corrosion. Useful in small doses, trouble in large ones — which is why we watch it.

TIP

A simple way to remember the three legs: **Carrier *levels*, Primary *brightens*, Wetter *un-pits*.** Three additives, three jobs.

REMEMBER

Brighteners get used up based on how much *plating* you do — not how many days go by. Track them by **amp-hours** (total current × time), not by the calendar. A busy day burns more brightener than a slow day, even though both are 24 hours long. Typical consumption runs roughly 0.5–2.0 mL carrier and 0.1–0.5 mL primary per 1,000 amp-hours, but **always go by your supplier’s TDS** — these numbers vary a lot between products.

HEADS-UP — THE BIG ONE FOR NEW OPERATORS

Do not add brightener on your own judgment. This is the single most common way a new hire wrecks a tank. Brighteners are dosed in milliliters into thousands of gallons — a heavy hand can over-brighten the bath in seconds and cause days of cracked, brittle, scrapped parts. Brightener additions are made by the plating engineer or your supervisor, guided by a Hull Cell test. Your job is to *notice* and *report*, not to dose.

• THE HULL CELL — YOUR CRYSTAL BALL

A **Hull Cell** is a tiny, trapezoid-shaped (slanted) plastic tank. Pour a small sample of your real bath into it, hang a little brass test panel in it at an angle, and plate for a few minutes under standard conditions. Because the panel sits at a slant, one end is close to the anode (high current) and the other is far (low current) — so a single panel shows what the bath does across the *entire* range of current densities at once. A common standard set is **267 mL, 2 amps, 10 minutes, brass panel** — but the current, time, and temperature are **supplier- and condition-dependent: run the Hull Cell at your bath's actual operating temperature** (130–150°F for this bath) and follow your brightener supplier's TDS.

WHAT THE PANEL LOOKS LIKE	WHAT IT'S PROBABLY TELLING YOU
Bright across the whole panel	Bath is in balance — happy days
Dull across the whole panel	Low Primary (Class II); or organic contamination
Bright at high CD, dull at low CD	Low Primary brightener
Bright at low CD, burnt at high CD	Too much brightener; OR low boric acid; OR pH too low
Hazy overall	Organic or metallic contamination
Dark streaks at low CD	Copper or lead contamination

💡 **TIP**

"CD" stands for **current density** — how much current hits each unit of area. High-CD spots are edges, points, and faces nearest the anode; low-CD spots are recesses, inside corners, and faces far from the anode. Almost every plating defect is really a CD story: "looks fine here, bad over there." Learn to think in high-CD vs low-CD and half the troubleshooting tables suddenly make sense.

• OPERATOR ESSENTIALS

TAPE A COPY NEAR THE TANK

PARAMETER	RACK	BARREL	NOTES
Temperature	130–150°F	130–150°F	Sweet spot 135–145°F
Nickel Sulfate ($NiSO_4 \cdot 6H_2O$)	240–330 g/L	270–330 g/L	Higher for barrel
Nickel Chloride ($NiCl_2 \cdot 6H_2O$)	30–60 g/L	45–60 g/L	Higher Cl helps anodes dissolve
Boric Acid (H_3BO_3)	30–45 g/L	37–45 g/L	Never below 30; crystallizes below ~100°F
pH	3.8–4.2	3.8–4.2	Down with H_2SO_4, up with $NiCO_3$
Current Density	20–60 ASF	5–15 ASF	Higher CD = faster, more burning risk
Cathode Efficiency	~95% (92–97%)	~95% (92–97%)	Supplier/condition-dependent; a little hydrogen always forms
Filtration	3–5 turnovers/hr	Continuous	Carbon canister handles organics
Anodes	S-Rounds (preferred)	S-Rounds in Ti baskets	S-Rounds are sulfur-depolarized (dissolve cleanly); R-Rounds vary — not always sulfur-activated
Anode-to-Cathode ratio	about 2:1	per basket setup	Enough anode area prevents passivation

Temperature too cold = dull, slow, won't level, boric acid drops out; too hot = wasted energy, degraded additives. In a hot shop in summer you may need *cooling*, not heating. **ASF** = amps per square foot; rack runs hot and fast (20–60), barrel runs gentle (5–15) because tumbling parts shadow each other. **Cathode efficiency** — typically ~95% (roughly 92–97%, and supplier-/condition-dependent) — means most current deposits nickel; the rest makes a little hydrogen gas.

⚙ TECHNICAL STUFF

That 3–8% of current that doesn't deposit nickel is making atomic hydrogen right at the steel surface. On ordinary mild steel, no problem. But on high-strength steel (above 40 HRC, or above 180 ksi tensile), that hydrogen can soak in and cause **hydrogen embrittlement**. We handle it with a post-plate bake (Station 06). For now: high efficiency does *not* mean zero hydrogen.

• WATTS BATH CROSS-SECTION

WATTS BATH CROSS-SECTION — TANK ANATOMY

• AGITATION, FILTRATION, AND ANODES — THE SUPPORTING CAST

AIR AGITATION

Oil-free air is bubbled through a pipe (the **sparger**) along the tank bottom. It keeps solution moving so fresh nickel is constantly delivered to the part, and sweeps away hydrogen bubbles before they pit the finish.

> ⚠ **HEADS-UP**
>
> It **must** be *oil-free* air. If your compressor leaks oil into the line, that oil goes straight into the bath and contaminates the organic additives — causing haze and pitting that can take a carbon treatment to fix. If you ever smell or see oil at the sparger, stop and report it.

FILTRATION (3–5 TURNOVERS/HR, CONTINUOUS)

A pump continuously sucks bath through filter bags to remove tiny solid particles that would make the finish rough. “3–5 turnovers per hour” means the entire tank volume passes through the filter three to five times every hour. A **carbon canister** can be plumbed in to pull dissolved organic junk out of solution.

> 💡 **TIP**
>
> Watch the filter pressure gauge. Slowly rising pressure means the bags are loading with dirt and need changing. A sudden spike or a drop in flow means trouble — clogged bag, closed valve, or failing pump. A clean, steady filter is a quiet hero of bright parts.

ANODES (S-ROUNDS IN TITANIUM BASKETS)

The nickel you plate has to come from somewhere — the **anodes**, nickel pieces in titanium mesh baskets on the positive side. As you plate, they dissolve to resupply the bath. We use **sulfur-depolarized** nickel (“S-Rounds”), which dissolve cleanly; plain electrolytic nickel dissolves sluggishly and can passivate. Baskets are wrapped in **anode bags** (polypropylene or cotton, usually doubled) that catch the fine sludge anodes shed. New bags must be soaked (“leached”) before first use.

> ★ **REMEMBER**
>
> Keep the **anode-to-cathode area ratio around 2:1** — roughly twice as much anode surface as part surface. Too little anode area and the anodes passivate (stop dissolving), the bath runs short on metal, and quality slides. If baskets are running low on nickel chunks, flag it.

• THE PROCEDURE — RUNNING A LOAD, STEP BY STEP

Start of shift — before any parts go in:

1. **Confirm temperature** in range (130–150°F, ideally 135–145°F). Never plate a cold tank.
2. **Check the pH** is 3.8–4.2. If drifting, do *not* adjust yourself unless authorized — flag it.
3. **Confirm agitation is running** — steady, even bubbling across the whole sparger, no dead zones.
4. **Check filtration** — pump running, pressure normal, flow good.
5. **Eyeball the bath** — clear and clean, not cloudy or foamy. Cloudy/foamy = do not run; report it.
6. A **Hull Cell** is typically run (by the engineer) to confirm brightness before production.
7. **Log your rectifier settings** for the load.

Running the load:

8. Lower the rack or load the barrel smoothly — no splashing (splashing throws nickel mist into the air).
9. Make sure the electrical connection to the cathode bar is solid. A loose connection means bare or under-plated spots.
10. Set the rectifier to the specified current for the load's total surface area, staying inside the ASF window.
11. Plate for the specified time. Thickness = current × time — more amp-hours, more nickel.
12. Watch and listen. Steady bubbling, steady current, no unusual smells.

End of the load:

13. Shut off current to that load before lifting, per your SOP.
14. Lift smoothly and let parts **drain back over the tank** a few seconds — returns valuable bath.
15. Move promptly to the post-plate rinse. Bright nickel oxidizes in air within seconds — don't let parts sit.
16. **Log the load** — settings, time, and anything unusual.

TIP — YOUR SENSES ARE INSTRUMENTS

A beginner learns the shop floor through their senses before they learn it through meters — so learn what *normal* looks, sounds, and smells like, and anything off will jump out at you. **A healthy nickel bath** is a clear, bright emerald-green and runs at a steady, even simmer of fine bubbles from the air agitation — not a rolling boil and not dead still. **It should smell faintly acidic, nothing harsh.** A **sharp, hot, "electrical" or acrid smell is burning** — current too high or a bad connection — back off and check. Cloudy or off-color solution, uneven or surging bubbling, a humming or arcing rectifier, or any new smell all mean "stop and look."

TIP

Get in the habit of a quick visual on every load as it comes out. You are the first quality inspector. A dull patch, a burnt edge, a haze, a pinhole field — catching it on load #1 saves you from running it on loads #2 through #20.

COMMON DEFECTS — WHAT YOU'LL SEE AND WHAT IT MEANS

You will see problems. Everybody does. The skill isn't avoiding every defect forever; it's *recognizing* them fast and knowing whether to fix, stop, or call for help.

WHAT YOU SEE	MOST LIKELY CAUSE	WHAT TO DO
Dull deposit (no shine)	Brightener low; organic contamination	Notify the engineer for a Hull Cell check
Burning at edges (rough, dark, powdery)	Current too high; pH or boric acid low	Reduce current; notify the engineer
Pitting (pinholes)	Low wetting agent; poor agitation	Check the air sparger; notify the engineer
Dark streaks	Copper or lead contamination	**Stop. Tag the tank. Notify the engineer.**
Peeling deposit	Poor cleaning upstream; high stress	Check the cleaning line; notify the engineer
Roughness (sandpaper feel)	Particulate; poor filtration; anode fines	Check filters; check anode bags
Haze (cloudy, not bright)	Iron or zinc contamination; organic breakdown	Notify the engineer; expect a metals analysis
Cracking	Too much carrier (Class I); stress; copper	Notify the engineer; do not add brightener
Skip plating (bare spots)	Passive oxide or soil from bad prep	Check cleaning and activation upstream

★ **REMEMBER**

Notice the pattern in that last column. For almost everything, the right move for an operator is **observe, stop if needed, and notify** — not improvise a chemical fix. The bath is a long-lived, expensive, finely-tuned system. The engineer makes the chemistry calls. Your eyes are the early-warning system that makes their job possible.

A WORD ON CONTAMINATION

The Watts bath can run for *years*, sometimes decades, without being dumped. The flip side: **every contaminant that gets in, stays in.** That's why Stations 1–4 obsess over rinsing. You don't need to memorize the chemistry, but know these limits exist — *dark streaks, haze, and milky deposits* are the visible symptoms:

METAL	MAX ALLOWED	HOW IT'S REMOVED (ENGINEER'S JOB)
Copper (Cu)	< 5 ppm	"Dummy plate" at low current (2–5 ASF)
Iron (Fe)	< 25 ppm	Raise pH + add H_2O_2, then filter
Zinc (Zn)	< 10 ppm	Dummy plate; or raise pH and filter
Chromium (Cr^{6+})	< 1 ppm	Reduce with sodium bisulfite, raise pH to precipitate, then filter
Lead (Pb)	< 0.5 ppm	Dummy plate; check anode purity
Cadmium (Cd)	< 0.5 ppm	Dummy plate; trace drag-back from legacy cadmium work (rare in modern shops — causes peeling/blistering)

TECHNICAL STUFF

"Dummy plating" means hanging a big sacrificial steel sheet in the bath and running it at very low current for hours. At low current density, the troublemaker metals (copper, lead, zinc) plate out onto the dummy preferentially, scrubbing them from solution while barely touching the nickel level. It's the bath's dialysis machine. "ppm" means parts per million — for lead, 0.5 ppm is half a gram in a thousand liters. Tiny amounts cause big problems, which is why upstream rinsing matters so much.

• SAFETY — THIS STATION DEMANDS YOUR RESPECT

Bright nickel plating is routine work, but “routine” is not the same as “harmless.” Read this carefully, then read it again next month.

⚠ HEADS-UP — HOT, ACIDIC SOLUTION

The bath runs at **130–150°F and pH 3.8–4.2** — hot and acidic enough to **burn your skin.** Wear full PPE every time: splash goggles *and* face shield, elbow-length chemical-resistant nitrile gloves, chemical apron, chemical-resistant boots. If you get bath on your skin, flush 15 minutes and report it.

⚠ HEADS-UP — NICKEL IS A SENSITIZER AND A CARCINOGEN BY INHALATION

Nickel compounds are classified as **known carcinogens by inhalation (GHS Category 1A)** and as **skin sensitizers.** “Sensitizer” means repeated skin contact can give you a permanent nickel allergy — **nickel itch** (allergic contact dermatitis). Once sensitized, it doesn’t go away. This is *cumulative and permanent*, so minimize skin contact starting on day one, even when exposure seems trivial. Gloves, every time.

⚠ HEADS-UP — NICKEL MIST IN THE AIR

The air agitation that keeps your finish bright also throws a fine **nickel-containing mist** above the tank — the inhalation hazard. **Local exhaust ventilation over the tank is mandatory**, and mist eliminators are recommended. Never lean your face over the tank without ventilation running. Limits: OSHA PEL **1.0 mg/m³** (as Ni) for soluble nickel compounds; the stricter ACGIH TLV is **0.1 mg/m³** for *soluble* nickel — and since this is a nickel-sulfate (soluble) bath, that 0.1 mg/m³ is the number to work to. If ventilation is down, the tank doesn’t run.

⚠ HEADS-UP — THE RECTIFIER IS HIGH AMPERAGE

That power supply pushes a *lot* of current. Before any maintenance, reaching into the tank, or work on bus bars or connections — **Lock Out / Tag Out (LOTO)** the rectifier. Physically lock the power off and tag it so nobody can switch it back on while your hands are in there. No exceptions, ever, no matter how quick the job seems.

⚠ HEADS-UP — RIGHT TANK, RIGHT METAL

Nickel sulfate solution corrodes mild steel. These tanks are built from **polypropylene (PP) or PVC plastic** for a reason. Don’t drop steel tools, fixtures, or hardware into the bath — they corrode, dump iron contamination, and can short across electrodes.

• TEACH-BACK CHECKLIST — STATION 05

- ☐ What are the cathode and anode, and which one is my part?
- ☐ Name the five core jobs in the bath (nickel sulfate, nickel chloride, boric acid, brighteners, pH) — what does each do?
- ☐ What's the operating window for **temperature** and **pH**, and which chemicals move pH up vs down?
- ☐ Why must boric acid never drop below 30 g/L, and what happens if the tank gets cold?
- ☐ Name the three brightener components and the defect each causes when *low* (carrier→poor leveling; primary→dull; wetter→pitting).
- ☐ What is a Hull Cell, and what does "bright at high CD, dull at low CD" tell you?
- ☐ What is "ASF," and why is barrel current density so much lower than rack?
- ☐ What does "3–5 turnovers per hour" mean, and why must the air be oil-free?
- ☐ Why does the anode-to-cathode ratio matter, and what is anode passivation?
- ☐ For dull, burning, pitting, dark streaks, peeling — what's the operator's correct response?
- ☐ What are the three big safety hazards here, and the right protection for each?
- ☐ Why must you **never** add brightener on your own?

• TOP MISTAKES TO AVOID — STATION 05

1. **Adding brightener yourself.** The #1 tank-wrecker. Notice and report; let the engineer dose.
2. **Plating a cold tank.** Cold bath plates dull, won't level, may have crystallized boric acid.
3. **Running cloudy or foamy bath.** If it doesn't look clean, don't run parts — report it.
4. **Letting parts sit in air after plating.** Bright nickel oxidizes in seconds; move to rinse promptly.
5. **Ignoring pitting.** Pinholes almost always mean low wetting agent or bad agitation — flag it.
6. **Reaching into the tank without LOTO.** High-amperage rectifier. Lock it out. Every time.
7. **Skipping gloves "just this once."** Nickel sensitization is permanent. There is no "just this once."
8. **Dropping steel into the bath.** Iron contamination is hard to remove and the tank never gets dumped.
9. **Not logging the load.** Your log is the trail that lets the engineer diagnose problems later.

SIGN-OFF — STATION 05

Trainee: ______________________ Trainer: ______________________ Date: ____________

"I can identify the cathode and anode, explain each bath component's function, state the operating windows for temperature, pH, boric acid, and current density, recognize the common defects and the correct operator response, and I understand the hot-acid, nickel-mist/sensitization, and high-amperage rectifier hazards — including LOTO before reaching into the tank. I understand brightener additions and chemistry adjustments are made only by authorized personnel."

STATION 06 OF 08

POST-PLATE RINSE & THE BAKE DECISION

THE QUIET STATION THAT SAVES REAL MONEY — AND DECIDES WHETHER A PART GOES TO THE OVEN

After the drama of the plating tank, the post-plate rinse feels like an afterthought. It is not. This little station does two big jobs: it protects your downstream chrome bath from being poisoned, and it's the decision point where you figure out whether a part needs to go to an *oven* before it goes anywhere else.

The analogy: imagine you just dipped a paintbrush in one bucket of paint and you're about to dip it in a second, different color. If you don't rinse the brush in between, you contaminate the second bucket. Every part coming out of the nickel tank is that loaded brush — dripping with nickel solution (that clinging film is **drag-out**). The post-plate rinse is the brush-cleaning step.

> ★ **REMEMBER**
> Nickel drag-out into the chrome tank is money down the drain *twice* — you lose the nickel you carried out, and you slowly ruin the chrome bath you carried it into. This station pays for itself.

• WHAT'S HAPPENING HERE, AND WHY

When your part leaves the nickel tank, it's coated in a thin film of hot nickel-sulfate-and-chloride solution. We rinse that off because: (1) **it protects the chrome bath** — nickel dragged into chrome contaminates it, shortens its life, and causes haze; (2) **it recovers valuable nickel** — that drag-out is dissolved money that can be captured and returned to the bath; (3) **it prevents spotting and haze** — leftover salts dry into spots, and slow transfer lets fresh nickel oxidize.

The measure of "clean enough" is **conductivity** (μS/cm) — a quick electrical measurement of how much dissolved stuff is in the water. Target: **below 200 μS/cm, goal under 100.**

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	`Ambient`	No heating needed
Time	`30–60 sec with agitation`	A double rinse is recommended
Conductivity	`< 200 μS/cm (target < 100)`	Nickel drag-out is the main concern
Drain time	`15–20 sec over the nickel tank`	This is how you recover drag-out
Drag-out recovery	`Optional stagnant first rinse`	Periodically returned to the nickel bath

DRAG-OUT RECOVERY — THREE WAYS TO CATCH THE NICKEL

METHOD	HOW IT WORKS	RECOVERY	BEST FOR
Stagnant recovery rinse	A still tank right after nickel; periodically pumped back to the bath	50–70%	Most rack and barrel lines
Spray rinse over the tank	DI sprayed on the rack while it drains back over the nickel tank	30–50%	High-value baths; tight floor space
Extended drain time	Just hold the part over the tank 15–20 sec before rinsing	20–30%	Every line — costs nothing

💡 **TIP**

Even if your line has no fancy recovery rinse, **the 15–20 second drain over the nickel tank is free and it works.** Letting the rack hang and drip back into the bath recovers 20–30% of the drag-out for zero cost. Build it into your muscle memory.

⚠ **HEADS-UP**

Don’t dump the first rinse. If your line has a stagnant recovery rinse, that first tank is *not* waste water — it’s concentrated nickel waiting to go home to the bath. Never drain it without checking; that’s literally pouring nickel down the drain.

THE BAKE DECISION — THE MOST IMPORTANT THING AT THIS STATION

Some steel parts must be **baked in an oven** after plating to prevent **hydrogen embrittlement**. The post-plate rinse is the decision point: a part that needs baking goes from here to the *oven* — not to the chrome tank. Lives can ride on this for safety-critical hardware.

What is hydrogen embrittlement? Plating (and acid activation back in Station 03) drives a little hydrogen into the steel surface. On ordinary mild steel, harmless. But on **hardened, high-strength steel**, that trapped hydrogen makes the metal brittle — so brittle it can crack later, suddenly, under load, sometimes weeks after it left your shop.

The cure is a bake. Heating drives the hydrogen back out. The governing spec is **ASTM B850**:

- **Who needs it:** Hardened steel above **40 HRC** (equivalently above about **180 ksi** tensile strength).
- **Temperature: 375 ± 25°F.**
- **Time:** A minimum of **8 hours** (longer for harder/higher-strength parts — often 8 to 24 hours).
- **Timing: Within 4 hours of the last electroplating step.**

⚠ **HEADS-UP — THE 4-HOUR CLOCK IS THE TRAP THAT CATCHES NEW OPERATORS**

The 4-hour window **starts the moment the part comes out of the last plating tank — nickel or chrome, whichever is last — not when it reaches the oven.** (For a part that gets chrome after nickel, the clock runs from the chrome; for some specs the bake is required *before* chrome, in which case it runs from the nickel — always check the spec.) If a baking-required part sits on a cart for three hours before someone loads the oven, you've burned three of your four hours just on transport. Delay past 4 hours may mean the bake *can no longer reverse the damage.* When in doubt, bake sooner, and never let a bake-required part sit.

THE BAKE DECISION IN PLAIN STEPS

1. **Check the work order / drawing.** Does the spec call for an H-embrittlement bake? Many do for hardware, fasteners, and anything safety-critical.
2. **Is the steel hard (over 40 HRC / over 180 ksi)?** If yes → **oven**, 375 ± 25°F, 8+ hours, within 4 hours of plating. ASTM B850.
3. **Medium-strength (over ~120 ksi / 830 MPa)?** Often still baked, typically 4–8 hours — follow the spec.
4. **Safety-critical part?** Bake per the drawing callout — check it, don't guess.
5. **None of the above and no spec callout?** No bake — proceed to post-treatment.
6. **Not sure? Ask your supervisor.** This is never a guess-it-yourself decision.

★ **REMEMBER**

Some specifications require the bake *before* chromium plating, not after. Always verify the bake timing against the customer specification — your supervisor or engineer will know. The part's destination after this rinse — oven or chrome tank — depends entirely on this call.

TRANSFER TIMING — MOVE IT ALONG

A second clock runs at this station, much shorter. **Bright nickel oxidizes within seconds of hitting air.** That fresh, reactive nickel grows an invisible **passive oxide film** if it sits — and chrome will not plate evenly over a passivated nickel surface, giving skip-plating and peeling chrome.

💡 **TIP**

For parts going to decorative chrome, keep the nickel-to-chrome transfer **under 60 seconds**, and keep parts *wet* the whole way. A wet part isn't oxidizing. Don't let racks sit dripping on a bench between nickel and chrome — that pause is where haze and skip-plate are born.

• THE PROCEDURE — STEP BY STEP

1. Lift the rack/barrel from the nickel tank and **hold it over the nickel tank to drain 15–20 seconds** — recovering drag-out.
2. **Check the work order for an H-embrittlement bake requirement.** This is your fork in the road.
3. Immerse in the rinse 30–60 seconds with agitation. Use **DI water** for the final rinse where possible — no mineral spots.
4. If a recovery rinse is in use, run the part through that *first*, then the flowing rinse.
5. **Route the part correctly:** Bake required → straight to the oven, mindful of the 4-hour clock. No bake, going to chrome → move fast, under 60 seconds, keep it wet.
6. Handle parts only with gloves — for your skin *and* for the finish.

• COMMON PROBLEMS — STATION 06

WHAT YOU SEE	MOST LIKELY CAUSE	WHAT TO DO
Chrome bath life shortening	Nickel drag-in from a weak rinse	Tighten the rinse; add a recovery rinse
Water spots on the nickel	Hard water; slow transfer	Switch to DI; minimize air exposure
Haze on the bright nickel	Surface oxidation during slow transfer	Reduce transfer time; keep parts wet
Nickel salt crystals on the rack	Inadequate rinsing; heavy drag-out	Extend the rinse; strip racks more often

⚠ **HEADS-UP — SAFETY, STATION 06**

It's still nickel. The rinse water carries **dilute nickel solution**, and nickel is a **skin sensitizer even at low concentrations.** Nickel itch is *cumulative and permanent* once you're sensitized — and it builds from exactly these "low-risk" exposures. **Wear gloves every time** you handle parts or racks; the dilute tank is the one that gets people. **Wet floors** are a slip hazard (wear non-slip footwear), and **the bake oven is hot** (375°F burn hazard) — use proper gloves and tools.

• TEACH-BACK CHECKLIST — STATION 06

- [ ] What is "drag-out," and why is nickel drag-out into the chrome tank a problem?
- [ ] What's the conductivity target, and what does conductivity actually measure?
- [ ] Name the three drag-out recovery methods and which one costs nothing.
- [ ] What is hydrogen embrittlement, and which parts are at risk (over 40 HRC / 180 ksi)?
- [ ] State the bake numbers: temperature, minimum time, deadline (375 ± 25°F, 8+ hr, within 4 hr).
- [ ] **When does the 4-hour clock start?** (When the part leaves the last plating tank — nickel or chrome, whichever is last — not the oven.)
- [ ] Why must the nickel-to-chrome transfer stay under 60 seconds?
- [ ] If unsure whether a part needs baking, what do you do?
- [ ] Why wear gloves even at a "dilute" rinse station?

• TOP MISTAKES TO AVOID — STATION 06

1. **Missing a required bake.** Always check the work order. A skipped bake on a critical part is a serious, potentially dangerous defect.
2. **Forgetting the bake clock starts at the tank, not the oven.** Don't let a bake-required part sit.
3. **Dumping the recovery rinse.** That's concentrated nickel meant to go back to the bath.
4. **Letting parts oxidize between nickel and chrome.** Under 60 seconds and keep them wet.
5. **Skipping the drain-over-tank.** That free 15–20 seconds recovers real money.
6. **Going gloveless because "it's just water."** Sensitization is permanent.

SIGN-OFF — STATION 06

Trainee: ______________________ Trainer: ______________________ Date: ____________

"I can explain drag-out and why nickel must not reach the chrome bath, state the conductivity target, and describe drag-out recovery. I understand hydrogen embrittlement and the ASTM B850 bake (375 ± 25°F, minimum 8 hours, within 4 hours of plating), that the 4-hour clock starts when the part leaves the plating tank, and that I check every work order for a bake and ask when unsure. I know dilute nickel rinse is still a skin sensitizer and gloves are required."

STATION 07 OF 08 · STAGE 8 FOLLOWS IN ITS OWN SECTION

POST TREATMENT — CHROME, SEAL, OR LACQUER

FINISHING THE SYSTEM — AND WHY BRIGHT NICKEL CAN'T STAND ALONE

You've made a beautiful, mirror-bright nickel finish. Surely that's the end? Not quite. Here's a truth that surprises most newcomers: **bright nickel, left bare, tarnishes and hazes within weeks** in most environments. The shine will dull on its own. The job isn't finished until you put a protective layer over the nickel — and *what* you put over it determines whether the finish lasts months or decades.

This **post-treatment** station is a fork in the road: decorative chromium, a clear lacquer, an advanced multilayer system, or — for some functional parts — nothing at all. Your job is to know each path, route parts correctly, and (especially with chrome) protect yourself, because this station has the most dangerous chemistry on the line.

★ **REMEMBER**

Think of the layers like a sandwich. The **nickel** underneath does the real corrosion protection — the thick, hardworking filling. The **chrome** on top is a thin, hard, blue-white slice that adds tarnish resistance, scratch resistance, and that bright "chrome look." The chrome is famous, but the nickel is the muscle.

☀ **FUN FACT**

That "chrome bumper" shine on a classic car? The chrome layer is astonishingly thin — often just **a quarter to three-quarters of a micron.** The nickel underneath is many times thicker. People credit the chrome for the protection, but it's really the nickel doing the heavy lifting, with the chrome as the good-looking, scratch-resistant face.

POST-TREATMENT OPTIONS — PICKING THE RIGHT PATH

TREATMENT	TYPICAL USE	THICKNESS	KEY PARAMETER
Decorative Chrome (Hex, Cr^{6+})	Auto trim, plumbing, hardware	0.25–0.75 µm	CrO_3 200–300 g/L; 95–125°F
Decorative Chrome (Tri, Cr^{3+})	RoHS-compliant applications	0.2–0.5 µm	Per supplier TDS; lower toxicity
Microporous Ni + Cr	High-corrosion duplex/triplex systems	System-dependent	Per ASTM B456; STEP test
Clear Lacquer	Indoor decorative	5–15 µm	Spray or dip; air/heat cure
Bare Bright Ni	Functional / solderable parts	N/A	Accept tarnish, or use VCI packaging

• POST-TREATMENT OPTIONS EXPLAINED

“Hex” vs “Tri” chrome: “Hex” is **hexavalent chromium (Cr^{6+})** — the traditional, high-performance chemistry, but a confirmed human carcinogen, heavily regulated and being phased out in many markets. “Tri” is **trivalent chromium (Cr^{3+})** — newer, far less toxic, used where **RoHS** (Restriction of Hazardous Substances, an EU regulation limiting substances like hex chrome) compliance is required. Tri is safer; hex still wins on some appearance/performance points. **Duplex/triplex + microporous chrome** is the heavy-duty corrosion system (multiple nickel layers plus chrome with microscopic pores that *spread out* corrosion attack so it can’t dig a deep pit), verified with the **STEP test** (ASTM B764) and CASS salt-spray. **Clear lacquer** is the simple route for indoor decorative parts. **Bare bright nickel** is fine for functional parts (e.g., solderable) protected with **VCI** (vapor corrosion inhibitor) packaging.

TIP

Operators don’t choose the post-treatment — the customer’s specification does, via the work order. Your job is to *read the work order* and route the part to the correct path. A part headed for chrome and a part headed for lacquer take very different routes and very different PPE.

DECORATIVE CHROME — THE EXACT NUMBERS

PARAMETER	HEX CHROME (CRO_3)	TRI CHROME (CR^{3+})
Concentration	CrO_3 200–300 g/L; CrO_3:SO_4 = 100:1	Cr^{3+} salt per supplier TDS
Temperature	95–125°F (35–52°C)	85–120°F (29–49°C)
Current Density	100–200 ASF	80–150 ASF
Time	3–8 min	3–10 min
Cathode Efficiency	10–18%	15–30%
Anodes	Lead-antimony (Pb-Sb)	Graphite / mixed metal oxide (MMO)

Two things stand out. **The current density is very high (100–200 ASF)** — far higher than nickel; chrome needs a hard electrical push to deposit at all. **The cathode efficiency is shockingly low (10–18% for hex).** Remember nickel was 92–97%? Chrome is the opposite — *most* of your electricity is splitting water and making gas; only a small fraction deposits chrome.

HEADS-UP

That low efficiency means chrome tanks generate a *huge* volume of gas bubbles, and those bubbles fling chrome-laden mist into the air. With hexavalent chrome, that mist is a carcinogen. This is exactly why a chrome tank requires mist suppressant and serious ventilation.

TECHNICAL STUFF

A hex chrome bath depends on a critical ratio between chromic acid (CrO_3) and its catalyst (sulfate), classically around 100:1. If that ratio drifts, the chrome comes out brown, yellow, hazy, or won't cover. This is delicate analytical chemistry managed by the engineer — operators don't adjust it, but you should recognize that "brown or yellow chrome" is a *ratio* symptom, not something you fix with more current.

TRANSFER TIMING STRIKES AGAIN

HEADS-UP

Keep the nickel-to-chrome transfer **under 60 seconds** and keep parts wet. If the nickel oxidizes (passivates) before reaching the chrome tank, the chrome will **skip-plate** (leave bare patches) or **peel.** Most "chrome won't cover" and "chrome peels" defects trace straight back to a part that sat too long between the two tanks.

REMEMBER

After post-treatment comes **Stage 8 (Final Rinse / Dry)** — the spot-free finish and pack-out. Because it is short, Stage 8 gets its own dedicated section right after this chapter, with its own Teach-Back and sign-off.

THE PROCEDURE — STEP BY STEP

1. **Read the work order** and identify the path: hex chrome, tri chrome, lacquer, or bare/functional.
2. For **chrome:** confirm full PPE is on (mandatory), transfer from nickel **under 60 seconds, kept wet**, plate per the parameters, then final rinse and dry.
3. For **lacquer:** apply by spray or dip per the product TDS, then **allow full cure (~24 hr air, or heat-cure per TDS) before packing.**
4. For **bare/functional:** confirm the spec accepts tarnish; protect with **VCI packaging** if specified.
5. **Final rinse** in DI, **dry** with warm air (120–140°F max).
6. **Handle with clean cotton/lint-free gloves** from here to packing.

• COMMON PROBLEMS — STATION 07

WHAT YOU SEE	MOST LIKELY CAUSE	WHAT TO DO
Chrome haze / milky look	Nickel drag-in; ratio off; temp too low	Improve post-Ni rinse; engineer checks CrO_3:SO_4 ratio and temp
Chrome skip-plate (bare patches)	Poor nickel quality; passivated nickel surface	Check nickel brightness; reduce transfer time
Brown / yellow chrome	Catalyst ratio off; contamination	Engineer analyzes and adjusts the bath
Chrome peels from nickel	Nickel oxidized; too long a delay	Cut Ni-to-Cr transfer to under 60 sec
Pitting in the chrome	Pits in the nickel showing through; poor wetting	Fix pitting at the nickel stage; add chrome wetting agent
Nickel tarnishes (no post-treat)	Expected — nickel isn't corrosion-proof alone	Apply post-treatment, or accept for functional use

★ **REMEMBER**

Most chrome defects are really *upstream* or *transfer* problems wearing a chrome costume. Haze and skip-plate usually mean nickel drag-in or a part that sat too long. Fix the rinse and the timing, and most chrome trouble disappears.

• SAFETY — STATION 07 — READ THIS TWICE

This station has the most hazardous chemistry on the entire line. Treat it with the seriousness it demands.

⚠ **HEADS-UP — HEXAVALENT CHROMIUM IS A CONFIRMED HUMAN CARCINOGEN**

Hexavalent chrome (Cr^{6+}) is classified by IARC as a **Group 1 carcinogen** — the highest, "confirmed in humans" category. The OSHA permissible exposure limit is extremely low: **5 µg/m³** (micrograms per cubic meter — *millionths* of a gram). This is not a chemical to be casual around.

⚠ **HEADS-UP — FULL PPE IS MANDATORY IN THE HEX CHROME AREA**

A **supplied-air respirator or PAPR** (powered air-purifying respirator) with **P100/OV cartridges**, a **full face shield**, a **chemical suit**, **double gloves**, and **chemical boots.** This is more PPE than anywhere else on the line, for good reason. Do not enter the chrome area, even to "just grab something," without proper protection.

⚠ **HEADS-UP — MIST SUPPRESSANT & VENTILATION ARE REQUIRED, NOT OPTIONAL**

Because chrome's low efficiency throws so much mist, a **chrome mist suppressant** must be maintained on hex chrome tanks and the **ventilation must be running.** If the suppressant is low or ventilation is down, the tank does not run.

⚠ **HEADS-UP — MEDICAL SURVEILLANCE & TRIVALENT CHROME**

Workers in the hex chrome area fall under **OSHA 29 CFR 1910.1026**, which requires medical monitoring — there to catch any exposure effects early. Take part in it. **Trivalent chrome (Cr^{3+})** is significantly less toxic than hex, but it is still industrial chemistry: standard chemical PPE and ventilation are still required.

• TEACH-BACK CHECKLIST — STATION 07

- [ ] Explain why bare bright nickel needs a post-treatment, and which layer in the “sandwich” does the real corrosion protection.
- [ ] State typical decorative chrome thickness (**0.25–0.75 µm**).
- [ ] Explain the difference between hex and tri chrome, and when tri is chosen.
- [ ] Explain why chrome’s efficiency is so low and why that creates a mist hazard.
- [ ] State why the Ni-to-Cr transfer must stay **under 60 seconds**, kept wet.
- [ ] State the rule on packing lacquered parts (full cure first, ~24 hr).
- [ ] List the full PPE required for the hex chrome area.
- [ ] State the OSHA limit and carcinogen category for hex chrome (**5 µg/m³; IARC Group 1**).

• TOP MISTAKES TO AVOID — STATION 07

1. **Entering the chrome area without full PPE.** No shortcuts, ever.
2. **Running a chrome tank with low mist suppressant or ventilation down.**
3. **Letting nickel oxidize before chrome.** Under 60 sec, kept wet.
4. **Packing lacquered parts before they cure.**
5. **Handling finished parts bare-handed.**
6. **Treating tri chrome as “harmless.”**
7. **Trying to fix brown/yellow chrome with current** — it’s a bath-ratio problem.

SIGN-OFF — STATION 07

Trainee: ______________________ Trainer: ______________________ Date: ____________

“I can explain why bright nickel requires post-treatment, identify the common options, and state decorative chrome thickness and operating windows. I understand the under-60-second nickel-to-chrome transfer and the final rinse, dry, and lacquer-cure rules. I understand hexavalent chromium is a confirmed human carcinogen (IARC Group 1, OSHA PEL 5 µg/m³), that full PPE including supplied-air/PAPR is mandatory in the hex chrome area, that mist suppressant and ventilation are required, and that medical surveillance applies under OSHA 29 CFR 1910.1026.”

STAGE 8 OF 08 · THE WRAP-UP

FINAL RINSE & DRY

THE LAST TEN SECONDS THAT CAN MAKE OR BREAK EIGHT STAGES OF WORK

Stage 8 is taught here, at the end of the Station 07 chapter, because it is short — but short is not the same as unimportant. After post-treatment, every part gets one last clean-up so it ships **spotless**. A single water spot, fingerprint, or skipped cure can reject a part that was perfect a minute earlier — and this is exactly where finished work is most often lost to carelessness, because operators relax once “the plating is done.”

★ **REMEMBER**

The finish you spent eight stages building can be ruined in the final ten seconds by a bare thumb or a rushed pack-out. Treat finished parts like glass: clean gloves, full cure, spot-free dry.

• WHAT STAGE 8 INVOLVES

- **DI (deionized) water rinse** — ambient, gentle immersion ~15–30 seconds — to remove the last traces of chemistry. DI prevents mineral water spots.
- **Warm-air dry** — **120–140°F maximum** — hot enough to dry quickly, gentle enough to avoid spotting and to protect lacquer.
- **Cure before packing** — chrome needs no cure (hardness is immediate); **lacquer must fully cure (~24 hr air, or heat-cure per the product TDS) before it is packed.**
- **Glove handling** — from the dryer to the box, handle only with **clean, dry, lint-free cotton or nitrile gloves.** Fingerprints on fresh chrome can become permanent.

• TEACH-BACK CHECKLIST — STAGE 8

- [ ] Explain why the final rinse uses **DI water** and what happens with hard water instead.
- [ ] State the **maximum dry temperature (120–140°F)** and why you don't dry hotter.
- [ ] State the rule for packing lacquered parts (**full cure first — ~24 hr air or heat-cure per TDS**) and that chrome needs no cure.
- [ ] Demonstrate correct glove handling of finished parts and explain why fingerprints on fresh chrome matter.

⚠ **HEADS-UP — SAFETY, STAGE 8**

The chemical hazard is low here, but the **dryer/oven is hot** (a burn hazard) and floors near the final rinse stay **wet and slippery.** Use proper gloves and tools at the dryer, wear non-slip footwear, and never pack a part that is still wet or uncured.

SIGN-OFF — STAGE 8 (FINAL RINSE / DRY)

Trainee: ____________________ Trainer: ____________________ Date: ____________

"I can run the final DI rinse and warm-air dry (120–140°F max) for a spot-free finish, I know lacquer must fully cure before packing while chrome needs no cure, and I handle finished parts only with clean lint-free gloves. I understand the hot-dryer burn and wet-floor slip hazards at this stage."

PART FOUR — BACK MATTER & CERTIFICATION
YOUR DECODER RING

GLOSSARY — EVERY TERM, PLAIN AND SIMPLE

ANYTIME A WORD IN THIS MANUAL MADE YOU PAUSE, FLIP BACK HERE

TIP
Keep this glossary tabbed. The fastest way to sound like a veteran on the shop floor is to use the right word for the right thing.

Acid activation (acid dip): A short dip in dilute acid (Stage 3) that dissolves the invisible oxide skin on metal so nickel can bond to clean, bare metal. Without it, nickel peels.
Adhesion: How well the plated layer sticks to the part. Adhesion failure = it flakes, blisters, or peels.
Agitation: Stirring or moving the bath (filtered air and/or moving parts). Keeps fresh chemistry against the surface and prevents pitting.
Ambient: Room temperature, roughly 60–85°F. No heater needed.
Amp-hour (Ah): Electrical "work" the bath has done — amps × hours. Brighteners are used up per amp-hour, not per day.
Anode: The positively charged metal (nickel "rounds") that dissolves into the bath to replace plated nickel. The part is the cathode.
Anode bag: A cloth sleeve over the anode that catches anode crumbs ("fines"). New bags must be soaked ("leached") before use.
ASF (amps per square foot): The unit for current density. Too low = dull; too high = burnt.
Boric acid (H_3BO_3): A mild acid (30–45 g/L) acting as a buffer — steadies pH at the part surface and prevents burning. Keep above 30 g/L.
Brightener: Organic chemicals that make the deposit shiny and level. Two classes (see below). The heart of bright nickel — and the fussiest part.
Buffer: A chemical that resists pH change. Boric acid is the bath's buffer.
Burning: Rough, dark, powdery deposit at high-current areas (edges, corners). From too much CD, low boric acid, low pH, or too hot.
Carbon treatment: Stirring activated carbon into the bath to soak up organic gunk, then filtering it out. The bath's "detox."
Carrier: Another name for the Class I brightener (the leveler).
Cathode: The part being plated — negatively charged, so positive nickel ions deposit on it.
Cathode efficiency: Percentage of current that deposits metal instead of making hydrogen gas. Bright nickel is high (typically ~95%, roughly 92–97%, supplier/condition-dependent); decorative chrome is low (10–18%).

• GLOSSARY (CONTINUED)

Class I brightener (carrier / leveler): Refines grain, levels the surface, co-deposits a little sulfur. Too much = brittle, peeling.

Class II brightener (primary): Delivers the mirror brightness across all current densities. Too little = dull everywhere.

Conductivity: A measure of dissolved salts/ions in rinse water (μS/cm). Low = clean rinse. How we prove a rinse is working.

Contaminant: Anything in the bath that doesn't belong — Cu, Fe, Zn, Cr, Pb, or organic breakdown. In nickel, contaminants accumulate and stay.

Counter-current rinse: A multi-tank rinse where clean water flows opposite to parts, so parts meet progressively cleaner water. Saves water, rinses better.

Current density (CD): See ASF. The amount of current per unit of part area.

DC (direct current): Steady, one-direction electric current (as opposed to alternating current). The rectifier converts shop AC into the DC that plating and electrocleaning require.

Deionized (DI) water: Water with dissolved minerals removed. Preferred for nickel lines.

Dendrite: A branching, tree-like metal growth defect, often from metallic contamination (copper, lead).

Drag-in: Unwanted chemistry a part carries *into* a tank from the previous tank (e.g., acid into nickel). The enemy.

Drag-out: Valuable chemistry a part carries *out* of a tank (e.g., nickel into the rinse). Costs money and pollutes downstream.

Drag-out recovery rinse: A still first rinse after plating that captures drag-out nickel to return to the bath.

Dummy plating (dummying): Plating onto a sacrificial cathode at low CD (2–5 ASF) to pull metallic contaminants out of the bath.

Ductility: How much the deposit can bend without cracking. Brightener imbalance and high stress reduce it.

Duplex / triplex nickel: A layered nickel system engineered so corrosion attacks the layers sideways instead of straight down to the steel — improving corrosion life.

• GLOSSARY (CONTINUED)

Electroclean (electrocleaning): Cleaning where DC current runs through an alkaline cleaner; the current generates scrubbing gas bubbles that strip the last microscopic film. Finish anodic to avoid metal smut.

Hull Cell: A small angled test tank that plates a single panel across a whole range of current densities at once. Reading the panel tells you what the bath needs. A common standard set: 267 mL, 2 A, 10 min, brass panel — current/time/temperature are supplier- and condition-dependent; run at your bath's actual operating temperature.

Hydrogen embrittlement (H-embrittlement): Atomic hydrogen soaks into high-strength steel during pickling and plating and makes it brittle, so it can crack later under load. Prevented by baking.

Inhibitor: A chemical added to acid that slows attack on the base metal while still removing oxide — and reduces hydrogen pickup. Used on high-strength steel.

LOTO (Lock-Out/Tag-Out): A safety procedure that physically locks off electrical power (the rectifier) before anyone reaches into a tank. Non-negotiable around live plating tanks.

Leveling: The brightener system's ability to fill in tiny scratches and tool marks, making a rough surface look smooth and bright.

Microsiemens per centimeter (µS/cm): The unit of conductivity. Lower = cleaner rinse water.

Mist eliminator: A device over a tank that captures harmful mist (nickel or chrome) before it reaches the air you breathe.

Nickel itch (allergic contact dermatitis): An allergic skin reaction from nickel exposure. **Cumulative and permanent once sensitized** — which is why we glove up every single time.

Oxide: A thin, invisible film of metal-plus-oxygen (tarnish/rust at a microscopic level) that forms on bare metal in air. Removed by acid activation before plating.

PEL (Permissible Exposure Limit): The OSHA legal limit for how much of a substance can be in the air over a work shift.

pH: A 0–14 scale of acidity (low) to alkalinity (high). 7 is neutral. The nickel bath runs slightly acidic at 3.8–4.2.

Pickle / pickling: Industry slang for the acid dip (acid activation). A "spent pickle" is acid that's worn out and full of dissolved metal.

Pitting: Small holes/pinholes in the deposit. Gas pitting comes from hydrogen bubbles sticking (low wetting agent / poor agitation); particulate pitting from debris.

PPE (Personal Protective Equipment): Your gear — goggles, face shield, gloves, apron, boots, respirator.

Rectifier: The machine that converts shop AC power into the DC current used for plating. High amperage — always LOTO before maintenance.

RoHS (Restriction of Hazardous Substances): An EU regulation limiting certain hazardous substances (including hexavalent chromium) in electrical and electronic products. RoHS-compliant work often drives the switch from hex to trivalent chrome.

S-Rounds / R-Rounds: Shapes of nickel anode. **S-Rounds** are sulfur-depolarized so they dissolve cleanly and are the preferred choice for the nickel bath. **R-Rounds** (rolled/electrolytic rounds) vary — they are not necessarily sulfur-activated, so they may dissolve less cleanly. Pure (electrolytic) nickel without a depolarizer dissolves poorly.

Skip plating: Bare, unplated spots where metal didn't take — usually from leftover oxide, fingerprints, or organic soil. A cleaning/activation failure.

Smut: A loose, dark film of redeposited metal or soil. Cathodic electrocleaning can cause it; an anodic finish removes it.

Strike: A thin, special first plating layer (e.g., copper or nickel strike) that promotes adhesion on tricky substrates like zinc die cast or brass.

Substrate: The base part you're plating onto — steel, brass, zinc die cast, etc.

Surfactant: A "surface-active agent" that lowers water's surface tension. The bath's wetting agent is a surfactant; cleaner surfactants can cause foaming in rinses.

Throwing power: The bath's ability to plate evenly into recesses and low-current areas, not just high spots. Nickel chloride and proper pH help.

Titrate / titration: A simple chemical test measuring how strong a solution is (e.g., cleaner or acid) so you know whether to add more.

TDS (Technical Data Sheet): The supplier's instruction sheet for a product. Always defer to the TDS.

UTS (ultimate tensile strength): The maximum tensile (pulling) stress a metal can take before it breaks, often in ksi (thousands of psi). High-strength steel (above ~180 ksi UTS / 40 HRC) is at risk of hydrogen embrittlement and needs a post-plate bake.

v/v (volume per volume): Concentration by volume, e.g., "HCl 10% v/v" = 10 parts acid per 100 parts solution.

Water-break test: The free cleanliness test. A clean part holds a continuous, unbroken sheet of water; a dirty (oily) part makes the water bead up and "break." Must pass before plating.

Watts bath: The classic, 80+-year-old bright nickel formula — nickel sulfate + nickel chloride + boric acid + brighteners. Named after Oliver Watts.

Wetting agent: The surfactant in the nickel bath that releases hydrogen bubbles from the surface so they don't leave pits. Low wetter = gas pitting.

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

★ **REMEMBER**

A defect is information, not a disaster. The parts are telling you exactly what the bath or the prep is missing. Learn to read them.

⚠ **HEADS-UP**

Before adjusting bath chemistry, confirm with a **Hull Cell** panel and your titration/analysis numbers. Guessing and dumping chemicals into a bath that runs for *years* can cause damage that takes weeks to undo. When in doubt, run a Hull Cell first.

CLEANING (STAGE 1)

SYMPTOM	LIKELY CAUSE	FIX
Water-break test fails (beads up)	Cleaner too weak or too cool	Titrate and boost cleaner; raise temp to 140–180°F
Oily film after cleaning	Floating oil redepositing; cleaner exhausted	Skim or dump the oil layer; recharge cleaner
White residue on parts	Hard-water salts; heavy cleaner drag-out	Check water quality; improve the rinse
Etching on zinc die cast	Cleaner too hot/alkaline for that substrate	Lower temp; switch to a mild cleaner for die cast
Skip plating downstream	Oxide film or fingerprints not removed	Add/extend anodic electroclean; handle with clean gloves

PRE-ACTIVATION RINSE (STAGE 2)

SYMPTOM	LIKELY CAUSE	FIX
High conductivity (>500 µS/cm)	Too much drag-out; low water flow	Drain longer over the cleaner; raise fresh-water flow
Soapy residue on parts	Rinse too short or too small	Extend dwell; add agitation; consider a double rinse
Acid dip turns alkaline fast	Cleaner carry-over overwhelming the acid	Tighten the rinse; add a second rinse stage
Foaming in the rinse	Surfactant carried from the cleaner	Increase water flow; review cleaner formulation

ACID ACTIVATION (STAGE 3)

SYMPTOM	LIKELY CAUSE	FIX
Dark oxide still on steel	Acid too dilute or dip too short	Increase concentration / time (watch H-embrittlement)
Pitting or etching of the part	Acid too strong, too long, or too warm	Reduce concentration; shorten dip; add inhibitor
Adhesion failure in the nickel	Incomplete activation; leftover oxide; alkaline drag-in	Re-check acid strength/time and the upstream rinse
Copper smut on brass	Dezincification — acid pulling zinc from the alloy	Shorten dip; use a cyanide copper strike before nickel
White powdery smut on die cast	Fluoride too strong or dip too long	Reduce concentration; shorten time; follow supplier TDS
Excessive HCl fuming	Too concentrated/hot; poor ventilation	Cut to minimum effective strength; check exhaust

PRE-PLATE (GUARD) RINSE (STAGE 4)

SYMPTOM	LIKELY CAUSE	FIX
Conductivity high (>200 µS/cm)	Heavy acid drag-out; low flow	Increase drain time off the acid dip; raise flow
Nickel bath pH drifting down	Acid drag-in from activation	Tighten conductivity target; add a second rinse
Iron climbing in the nickel bath	Dissolved Fe dragged in from spent pickle	Refresh the pickle; improve rinse; add a stage
Chloride climbing in the nickel bath	HCl drag-in	Switch activation to H_2SO_4; or improve rinse
Parts oxidize before plating	Transfer too slow; parts drying	Speed transfer — enter nickel within ~30 sec

BRIGHT NICKEL PLATING (STAGE 5)

SYMPTOM	LIKELY CAUSE	FIX
Burning at high CD (edges/corners)	pH too low; boric acid low; too hot; excess brightener	Check pH, boric acid, temp; Hull Cell; reduce CD
Dull deposit everywhere	Class II depleted; too cold; organic contamination	Add Class II per Ah; check temp; carbon treat
Gas pitting (pinholes)	Low wetting agent; poor agitation; organics	Add wetter; boost air; carbon treat; check filter
Haze across the panel	Iron or zinc contamination; organic breakdown	Analyze metals; carbon treat; dummy plate
Dark streaks/spots at low CD	Copper or lead contamination; anode sludge	Analyze Cu/Pb; dummy plate at 2–5 ASF; check anode bags
Milky/white haze	Zinc contamination; excess carrier	Analyze Zn; dummy plate; cut carrier addition rate
Roughness (sandpaper feel)	Particulate; poor filtration; anode fines	Check filters; replace anode bags; raise filtration rate
Peeling / poor adhesion	Bad cleaning; incomplete activation; excessive stress	Re-check water-break and acid dip; measure deposit stress
Cracking (fails 180° bend)	Excess Class I; high stress; too thick; Cu contamination	Reduce Class I; carbon treat; check Cu; Hull Cell bend test
Poor throwing power (recesses dull)	Low nickel chloride; pH too low; too cold	Raise $NiCl_2$ to 50–60 g/L; optimize pH; check temp
Skip plating (bare areas)	Passive oxide; organic soil; weak electroclean	Improve cleaning; add anodic electroclean; check activation

TECHNICAL STUFF

When you raise pH to 5.0–5.5 and add 3% v/v hydrogen peroxide during a carbon treatment, you're oxidizing dissolved iron from Fe^{2+} (which stays in solution) to Fe^{3+} (which precipitates as a solid you can filter out). A neat bit of redox chemistry that physically removes the iron — then bring pH back to 3.8–4.2 with dilute sulfuric acid.

POST-PLATE RINSE (STAGE 6)

SYMPTOM	LIKELY CAUSE	FIX
Chrome bath life unusually short	Nickel drag-in	Tighten the post-plate rinse; add a recovery rinse
Water spots on the nickel	Hard-water minerals; slow transfer	Use DI; minimize air exposure time
Haze appearing on bright nickel	Oxidation during a slow transfer	Speed up transfer; keep parts wet
Nickel salt crystallizing on racks	Inadequate rinsing; high drag-out	Extend rinse; strip racks more often

POST TREATMENT (STAGE 7)

SYMPTOM	LIKELY CAUSE	FIX
Chrome haze / milky look	Nickel drag-in; ratio off; too cold	Improve post-Ni rinse; check CrO_3:SO_4 ratio; check temp
Chrome won't cover (skip plate)	Poor or passive nickel surface underneath	Check Ni brightness; cut transfer time; check activation
Brown or yellow chrome	CrO_3-to-catalyst ratio off; contamination	Analyze bath; adjust ratio; check for metallic contamination
Chrome peels from the nickel	Poor Ni adhesion; oxidized Ni; too much delay	Cut Ni-to-Cr transfer to <60 sec; verify Ni adhesion
Pitting in the chrome	Pits in the nickel showing through; poor wetting	Fix pitting at the nickel stage; add chrome wetting agent
Bare nickel tarnishes over time	Normal — bright Ni isn't corrosion-proof alone	Apply post-treatment, or spec parts that accept tarnish

TIP

Notice how many "chrome problems" trace back to the nickel stage. Most downstream defects are born upstream. Fix the root, not the symptom.

TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO

★ **REMEMBER**

No solo work at a station until the **Trainer sign-off** AND the **Operator sign-off** boxes are both filled and dated. Training without a signature didn't happen, as far as an audit is concerned.

Competency levels: O = Observed (watched a trained operator) · **A** = Assisted (performed under direct supervision) · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT DUE
1	Stage 1 — Soak Clean / Electroclean	____	__________	__________	______
2	Water-break test (perform & interpret)	____	__________	__________	______
3	Stage 2 — Pre-Activation Rinse + conductivity	____	__________	__________	______
4	Stage 3 — Acid Activation (substrate-specific)	____	__________	__________	______
5	Stage 4 — Pre-Plate (Guard) Rinse	____	__________	__________	______
6	Stage 5 — Bright Nickel Plating operation	____	__________	__________	______
7	Bath maintenance — additions per amp-hour	____	__________	__________	______
8	Hull Cell — run & interpret panel	____	__________	__________	______
9	Carbon treatment / dummy plating	____	__________	__________	______
10	Stage 6 — Post-Plate Rinse + drag-out recovery	____	__________	__________	______
11	H-embrittlement bake decision & logging	____	__________	__________	______
12	Stage 7 — Post Treatment (Cr / seal / lacquer)	____	__________	__________	______
13	Stage 8 — Final Rinse / Dry & inspection	____	__________	__________	______
14	PPE selection & donning (all stations)	____	__________	__________	______
15	LOTO on rectifier	____	__________	__________	______
16	Emergency response — eyewash, shower, spill, first aid	____	__________	__________	______
17	SDS location & reading	____	__________	__________	______
18	Troubleshooting — symptom recognition	____	__________	__________	______

Operator name: ______________________________ **Hire/start date:** ____________________

Supervisor name: ______________________________ **Review cycle:** ☐ Annual ☐ Other: ______________

⚠ **HEADS-UP**

Safety rows (14–17) are not optional and not "fill in later." An operator may not work *any* station until PPE, LOTO, and emergency response are signed off. Safety competency comes before production competency, always.

💡 **TIP**

Set a recurring re-certification date (annual is common). People drift from procedure over time, and a yearly refresher catches bad habits before they become defects — or injuries.

20 QUESTIONS · PASSING SCORE 80% (16/20)

BRIGHT NICKEL COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED

★ **REMEMBER**

Passing score: 80% (16 of 20 correct). Anything safety-related (Questions 16–20) must be answered correctly to certify, regardless of total score. **Grader note:** a miss on any of Q16–Q20 is an automatic fail of the safety gate — re-teach and re-test before sign-off, no exceptions. The Answer Key follows — no peeking until you're done!

Name: ______________________ **Date:** ______________ **Score:** ________ / 20

Q1. The water-break test passes when:
A. Water forms a continuous, unbroken sheet   B. Water beads up into droplets   C. The part is dry within 5 seconds   D. The rinse water turns cloudy

Q2. During electrocleaning, why do you finish the cycle *anodic* (last 15–30 seconds)?
A. To heat the part faster   B. To deposit a protective layer   C. To strip off metallic smut that cathodic cleaning can deposit   D. To save electricity

Q3. (Short answer) What is the purpose of Stage 3, the acid activation (acid dip)? In one or two sentences, explain what it removes and why that matters for the nickel.

Q4. The pre-plate (guard) rinse, Stage 4, targets a conductivity of:
A. < 5,000 µS/cm   B. < 2,000 µS/cm   C. < 500 µS/cm   D. < 200 µS/cm (target < 100)

Q5. (Short answer) Why is the pre-plate rinse often called "the single most important contamination barrier on the line"? (Hint: how long does a nickel bath last, and what happens to contaminants in it?)

Q6. A standard Watts bright nickel bath contains all of the following EXCEPT:
A. Nickel sulfate   B. Nickel chloride   C. Boric acid   D. Sodium hydroxide (NaOH) for pH control

Q7. (Short answer) Name the correct operating ranges for the bright nickel bath's **temperature** and **pH**.

Q8. Boric acid in the nickel bath functions as a:
A. pH buffer that prevents burning   B. Brightener   C. Wetting agent   D. Metal source

Q9. An operator sees a *dull deposit across the entire panel* with no response to brightener additions. The most likely next step is:
A. Raise current density to 200 ASF   B. Add more nickel chloride   C. Carbon treat the bath (organic contamination suspected)   D. Lower temperature to ambient

Q10. (Short answer) What is the difference between Class I (carrier) and Class II (primary) brighteners — what does each do for the deposit?

Q11. “Burning” at high-current-density areas can be caused by all of the following EXCEPT:
A. pH too low   B. Boric acid too low   C. Too much wetting agent   D. Temperature too high

Q12. A cleaned part fails the water-break test (water beads up). The correct response is:
A. Proceed anyway — it looks shiny   B. Add more brightener to the nickel bath   C. Skip straight to the nickel tank   D. Stop and re-clean the part before continuing

Q13. (Short answer) What is a Hull Cell, and why is it the plater’s most important diagnostic tool? Name the standard test conditions if you can.

Q14. Why run a *drag-out recovery rinse* right after the nickel bath (Stage 6)?
A. To capture drag-out nickel for return to the bath, saving money and protecting the chrome tank   B. To cool the parts before chrome   C. To raise the pH of the parts   D. It is purely decorative

Q15. (Short answer) In Stage 7, decorative chrome over bright nickel is very thin (0.25–0.75 µm). If the chrome is so thin, what is the nickel underneath actually doing for the part?

⚠ **SAFETY SECTION — ALL FIVE MUST BE CORRECT TO CERTIFY**

Questions 16–20 cover hazards that can permanently harm you or others. A miss here means re-teach and re-test before sign-off.

Q16. Nickel compounds are classified as:
A. Completely non-hazardous   B. Carcinogens (GHS Category 1A by inhalation) and skin sensitizers   C. Flammable liquids only   D. Safe to handle with bare hands

Q17. (Short answer) What is “nickel itch,” and why do we wear gloves *every* time we handle nickel-wet parts or racks — even at low concentration?

Q18. Before reaching into a plating tank for maintenance, you must:
A. Turn the lights off   B. Lock-Out/Tag-Out (LOTO) the rectifier   C. Lower the bath temperature   D. Notify a coworker by text

Q19. (Short answer) Hexavalent chromium (Cr^{6+}) is one of the most dangerous chemicals on the line. State (a) what health hazard it poses, and (b) one PPE/engineering control required when working the hex chrome tank.

Q20. For hardened steel parts (>40 HRC), hydrogen embrittlement relief requires baking at:
A. 212°F for 1 hour, any time that week   B. 375 ±25°F for at least 8 hours, started within 4 hours of the last plating step   C. 500°F for 30 minutes, the next day   D. No baking is ever required

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD

Q1. A — a continuous, unbroken water sheet. Beading means oily contamination remains.

Q2. C — to strip off metallic smut. Cathodic cleaning makes more scrubbing gas but can redeposit metallic smut; an anodic finish removes it.

Q3. It dissolves the **invisible surface oxide film** that forms during cleaning and rinsing, exposing **bare, chemically active metal** so the nickel can **bond/adhere**. Under-dip leaves oxide and causes adhesion failure; over-dip etches and loads hydrogen.

Q4. D — < 200 µS/cm, target < 100.

Q5. Because the **nickel bath runs hot (130–150°F) and is essentially never dumped — it can run for years or decades.** Every milligram of acid, iron, copper, or zinc that gets through **accumulates permanently**, so stopping contamination *before* it enters is critical.

Q6. D — never use NaOH to raise pH; sodium doesn't belong in the bath. Use nickel carbonate / nickel hydroxide to raise pH and sulfuric acid to lower it.

Q7. Temperature 130–150°F (54–66°C) (target ~135–145°F) and **pH 3.8–4.2.**

Q8. A — a pH buffer at the cathode film that prevents burning. Must stay above 30 g/L (range 30–45 g/L).

Q9. C — carbon treat the bath. A dull panel that won't respond to Class II additions points to **organic contamination.**

Q10. Class I (carrier/leveler): refines grain, provides **leveling**, co-deposits sulfur; too much causes stress/brittleness. **Class II (primary):** provides the **mirror brightness** across the full current-density range; too little causes overall dullness.

Q11. C — too much wetting agent does *not* cause burning (it causes foam/haze). Burning comes from **low pH, low boric acid, high temperature, excess brightener, or excessive current density.**

Q12. D — stop and re-clean. A water-break failure means oily contamination remains; plating over it causes pitting, skip plating, and peeling. Shiny is not the same as clean.

Q13. A **Hull Cell** is a small angled test tank that plates one panel across a **whole range of current densities at once**, so the pattern reveals what the bath needs. It's the plater's diagnostic "microscope." A common standard set: **267 mL, 2 A, 10 minutes, brass panel** (current/time/temperature are supplier- and condition-dependent — run at the bath's actual operating temperature).

Q14. A — to capture drag-out nickel for return to the bath, saving money and protecting the downstream chrome tank from nickel contamination.

Q15. The **nickel does the real corrosion protection** and provides the bright, leveled, mirror surface. The thin chrome only adds tarnish resistance, scratch resistance, and the blue-white chrome appearance — the nickel underneath does the heavy lifting.

⚠ SAFETY ANSWERS (16–20) — ALL MUST BE CORRECT TO CERTIFY

Q16. B — carcinogens (GHS Category 1A by inhalation) and skin sensitizers.

Q17. Nickel itch is **allergic contact dermatitis** caused by nickel exposure. We glove up every time because the sensitization is **cumulative and permanent once it develops** — even dilute, low-concentration nickel solution can sensitize you over time.

Q18. B — Lock-Out/Tag-Out (LOTO) the rectifier. Live DC current in solution is deadly.

Q19. (a) Hexavalent chromium is a **confirmed human carcinogen (IARC Group 1)**; OSHA PEL 5 µg/m³ as Cr(VI). (b) Any one of: **supplied-air respirator or PAPR with P100/OV cartridge, full face shield, chemical suit, double gloves, chemical boots, mandatory chrome mist suppressant on the tank, local exhaust ventilation**, plus medical surveillance per OSHA 29 CFR 1910.1026.

Q20. B — 375 ±25°F for a minimum of 8 hours, started within 4 hours of the last electrodeposition step (per ASTM B850).

Answer distribution (MC): A=3 · B=3 · C=3 · D=3

★ REMEMBER

A score of 16/20 passes — but every safety question (16–20) must be correct to certify. If a trainee misses a safety item, re-teach it and re-test before sign-off. We don't gamble with people.

PP

PLATING POSTERS

PLATING POSTERS INC · METAL FINISHING REFERENCE SERIES

CERTIFICATE OF COMPLETION

BRIGHT NICKEL PLATING TRAINING PROGRAM

This certifies that

____

OPERATOR NAME

has successfully completed the **Bright Nickel Plating** training course — covering all eight process stages (Clean, Rinse, Activate, Rinse, Plate, Rinse, Post-Treat, and Final Rinse/Dry), bath chemistry and maintenance, Hull Cell diagnostics, troubleshooting, and the safety practices for alkaline cleaners, mineral acids, the nickel bath, and hexavalent chromium — and has passed the Completion Test with a score of ________ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: ____________________ **Re-certification due:** ____________________

____
TRAINEE SIGNATURE

____
CERTIFYING TRAINER

____
SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer specifications, and applicable regulatory requirements before production use. Nickel compounds are classified as carcinogens by inhalation and known skin sensitizers; hexavalent chromium is a confirmed human carcinogen — always consult current SDS and comply with OSHA, EPA, REACH, and local regulations.